



TRIBHUVAN UNIVERSITY  
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Department of Computer Science and Information Technology  
Patan Multiple Campus, Lalitpur

*A PROJECT REPORT*  
**NYAYAK: DIGITAL LEGAL NAVIGATION SYSTEM  
DESIGNED FOR NEPAL**

Submitted to:  
Department of Computer Science and Information Technology  
Patan Multiple Campus, Lalitpur

*In partial fulfillment of the requirements for the degree of Bachelor of Science in Computer  
Science and Information Technology  
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## **SUPERVISOR RECOMMENDATION**

I hereby recommend that this project prepared under my supervision by **Samir Paudel** and **Prakashman Singh Thakuri**, entitled “NYAYAK: DIGITAL LEGAL NAVIGATION SYSTEM DESIGNED FOR NEPAL,” in partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Information Technology, be processed for evaluation.

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## **APPROVAL LETTER**

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# ABSTRACT

Nyayak is a full-stack digital legal navigation platform designed for Nepal’s legal corpus and court infrastructure. The system addresses three persistent barriers to justice: limited public understanding of rights under the Constitution of Nepal [1] and the Civil Code 2017 [2], weak digital visibility for qualified lawyers outside major urban centres, and the lack of structured court information across Nepal’s 77 districts. It integrates an AI legal assistant, a lawyer marketplace, a community forum, and a court navigation module within a single web platform built on Next.js, Convex, and FastAPI. The legal assistant uses a hybrid RAG pipeline that combines BM25 retrieval with sentence-transformer semantic search, a design supported by prior work on RAG and hybrid retrieval [3–5].

The lawyer marketplace applies content-based recommendation to rank suitable legal professionals [6, 7], while the forum uses a PageRank-inspired influence model [8] and the court navigator uses shortest-path routing over a weighted district-level graph [9]. Evaluation on 50 legal queries achieved an 86% source-citation relevance rate, average latency of 3.2 seconds, and full success across eight system test cases. Deployed on free-tier infrastructure, Nyayak demonstrates that modern retrieval, recommendation, and graph-based methods can support an accessible and practical digital legal guidance platform for Nepal.

**Keywords:** Retrieval-Augmented Generation, Legal AI, Nepal Constitution, Hybrid Search, BM25, Semantic Similarity, PageRank, Dijkstra’s Algorithm, Content-Based Filtering, Full-Stack Web Application, Next.js, FastAPI, Convex, Access to Justice

# TABLE OF CONTENTS

|                                                                |             |
|----------------------------------------------------------------|-------------|
| <b>TITLE PAGE</b>                                              | <b>1</b>    |
| <b>SUPERVISOR RECOMMENDATION</b>                               | <b>i</b>    |
| <b>APPROVAL LETTER</b>                                         | <b>ii</b>   |
| <b>ACKNOWLEDGEMENT</b>                                         | <b>iii</b>  |
| <b>ABSTRACT</b>                                                | <b>iv</b>   |
| <b>TABLE OF CONTENTS</b>                                       | <b>vi</b>   |
| <b>LIST OF ABBREVIATIONS</b>                                   | <b>vii</b>  |
| <b>LIST OF FIGURES</b>                                         | <b>viii</b> |
| <b>LIST OF TABLES</b>                                          | <b>ix</b>   |
| <b>1 Introduction</b>                                          | <b>1</b>    |
| 1.1 Introduction . . . . .                                     | 1           |
| 1.2 Problem Statement . . . . .                                | 2           |
| 1.3 Objectives . . . . .                                       | 3           |
| 1.4 Scope and Limitations . . . . .                            | 4           |
| 1.5 Development Methodology . . . . .                          | 4           |
| 1.6 Report Organisation . . . . .                              | 5           |
| <b>2 Background Study</b>                                      | <b>5</b>    |
| 2.1 Background Study . . . . .                                 | 5           |
| 2.1.1 Retrieval-Augmented Generation (RAG) . . . . .           | 5           |
| 2.1.2 BM25 Probabilistic Ranking . . . . .                     | 6           |
| 2.1.3 Semantic Similarity with Sentence Transformers . . . . . | 7           |
| 2.1.4 Content-Based Filtering for Recommendation . . . . .     | 8           |
| 2.1.5 PageRank and Graph-Based Influence Scoring . . . . .     | 8           |
| 2.1.6 Dijkstra's Shortest-Path Algorithm . . . . .             | 9           |
| 2.2 Literature Review . . . . .                                | 9           |
| 2.2.1 Related Work in Legal AI . . . . .                       | 9           |
| <b>3 System Analysis</b>                                       | <b>11</b>   |
| 3.1 System Analysis . . . . .                                  | 11          |

|          |                                                                 |           |
|----------|-----------------------------------------------------------------|-----------|
| 3.1.1    | Requirement Analysis . . . . .                                  | 11        |
| 3.1.2    | Use Case Diagram . . . . .                                      | 14        |
| 3.1.3    | Feasibility Analysis . . . . .                                  | 15        |
| 3.1.4    | Object-Oriented Analysis . . . . .                              | 18        |
| <b>4</b> | <b>System Design</b>                                            | <b>20</b> |
| 4.1      | System Architecture . . . . .                                   | 20        |
| 4.2      | Database Design . . . . .                                       | 22        |
| 4.3      | Algorithm Details . . . . .                                     | 22        |
| 4.3.1    | Algorithm 1: Hybrid Retrieval-Augmented Generation (RAG) . . .  | 22        |
| 4.3.2    | Algorithm 2: Content-Based Lawyer Recommendation . . . . .      | 23        |
| 4.3.3    | Algorithm 3: Graph-Based Influence and DFS Reconstruction . . . | 23        |
| 4.3.4    | Algorithm 4: Jurisdictional Pathfinding . . . . .               | 24        |
| 4.4      | Component Diagram . . . . .                                     | 24        |
| <b>5</b> | <b>Implementation And Testing</b>                               | <b>25</b> |
| 5.1      | Implementation . . . . .                                        | 25        |
| 5.1.1    | Tools and Technologies Used . . . . .                           | 25        |
| 5.1.2    | Implementation Details of Modules . . . . .                     | 25        |
| 5.2      | Testing . . . . .                                               | 30        |
| 5.2.1    | Unit Test Cases . . . . .                                       | 30        |
| 5.2.2    | System Test Cases . . . . .                                     | 30        |
| 5.3      | Result Analysis . . . . .                                       | 30        |
| <b>6</b> | <b>Conclusion And Future Recommendations</b>                    | <b>31</b> |
| 6.1      | Conclusion . . . . .                                            | 31        |
| 6.2      | Future Recommendations . . . . .                                | 32        |
|          | <b>REFERENCES</b>                                               | <b>34</b> |
| <b>A</b> | <b>Screenshots of the System</b>                                | <b>37</b> |
| <b>B</b> | <b>Major Source Code Components</b>                             | <b>44</b> |
| <b>C</b> | <b>Project Progress Log Report</b>                              | <b>50</b> |

## LIST OF ABBREVIATIONS

| Abbreviation | Full Form                                                 |
|--------------|-----------------------------------------------------------|
| AI           | Artificial Intelligence                                   |
| API          | Application Programming Interface                         |
| BM25         | Best Match 25 (Probabilistic Ranking Function)            |
| CORS         | Cross-Origin Resource Sharing                             |
| CSIT         | Computer Science and Information Technology               |
| DFS          | Depth-First Search                                        |
| ER           | Entity Relationship                                       |
| FastAPI      | Fast Application Programming Interface (Python framework) |
| HMAC         | Hash-based Message Authentication Code                    |
| HTTP         | Hypertext Transfer Protocol                               |
| IDF          | Inverse Document Frequency                                |
| JWT          | JSON Web Token                                            |
| LLM          | Large Language Model                                      |
| MAU          | Monthly Active Users                                      |
| NLP          | Natural Language Processing                               |
| NPR          | Nepalese Rupee                                            |
| RAG          | Retrieval-Augmented Generation                            |
| REST         | Representational State Transfer                           |
| SHA          | Secure Hash Algorithm                                     |
| SSR          | Server-Side Rendering                                     |
| TF           | Term Frequency                                            |
| TF-IDF       | Term Frequency-Inverse Document Frequency                 |
| TU           | Tribhuvan University                                      |
| UI           | User Interface                                            |
| UX           | User Experience                                           |

## List of Figures

|      |                                                              |    |
|------|--------------------------------------------------------------|----|
| 3.1  | Nyayak Use Case Diagram . . . . .                            | 15 |
| 3.2  | Project Gantt Chart - 14-Week Development Schedule . . . . . | 17 |
| 3.3  | Nyayak Object Diagram . . . . .                              | 18 |
| 3.4  | Nyayak Activity Diagram - Legal Query Flow . . . . .         | 19 |
| 3.5  | Nyayak Sequence Diagram - Legal Query Session . . . . .      | 20 |
| 4.1  | Nyayak Three-Tier System Architecture . . . . .              | 21 |
| 4.2  | Nyayak Component Diagram . . . . .                           | 24 |
| A.1  | Landing Page - Desktop . . . . .                             | 37 |
| A.2  | Clerk Authentication Signup Page . . . . .                   | 38 |
| A.3  | Landing Page - Mobile and Navigation Drawer . . . . .        | 39 |
| A.4  | AI Chat Interface with Lawyer Recommendation Panel . . . . . | 39 |
| A.5  | Lawyer Directory - Find Lawyer Section . . . . .             | 40 |
| A.6  | Pending Cases - Lawyer Docket View . . . . .                 | 40 |
| A.7  | Lawyer Edit Profile Section . . . . .                        | 41 |
| A.8  | Citizen Dashboard . . . . .                                  | 41 |
| A.9  | Lawyer Dashboard . . . . .                                   | 42 |
| A.10 | Community Forum - Nested Threads . . . . .                   | 42 |
| A.11 | Court Navigation Map . . . . .                               | 43 |
| A.12 | FAQ Section . . . . .                                        | 43 |
| A.13 | Footer Section . . . . .                                     | 44 |

## List of Tables

|     |                                                                 |    |
|-----|-----------------------------------------------------------------|----|
| 3.1 | Functional Requirements Summary . . . . .                       | 13 |
| 3.2 | Non-Functional Requirements Summary . . . . .                   | 14 |
| 3.3 | Technical Feasibility Assessment . . . . .                      | 16 |
| 3.4 | Project Cost Breakdown (Economic Feasibility) . . . . .         | 17 |
| 4.1 | Convex Database Schema - Collections and Indexes . . . . .      | 22 |
| 5.1 | Tools and Technologies Used . . . . .                           | 26 |
| 5.2 | Unit Test Cases - Core Modules . . . . .                        | 28 |
| 5.3 | System Test Cases . . . . .                                     | 29 |
| 5.4 | RAG Evaluation Results and System Performance Summary . . . . . | 30 |

# 1. Introduction

## 1.1. Introduction

Access to legal services and information remains one of the most significant and least-addressed barriers faced by citizens across developing nations. Nepal is no exception. The Nepali legal system - governed by a comprehensive Constitution ratified in 2015 (2072 B.S.) [1], supplemented by the Civil Code 2017 [2], the Labour Act 2074, the Land Act 2021, and dozens of provincial statutes - is complex, multilingual, and largely inaccessible to ordinary citizens without professional legal counsel. With more than 150,000 cases pending in courts across the country, the gap between those who can effectively navigate the justice system and those who cannot continues to widen.

The access-to-justice deficit in South Asia is well-documented in academic literature. Information asymmetry - not merely cost - has been identified as a major barrier to legal participation, as citizens who lack basic awareness of their legal rights are structurally excluded from the justice system regardless of formal legal guarantees. This challenge is directly applicable to Nepal, where the Constitution guarantees equality before the law and the right to legal representation, yet no reliable digital resource exists to translate those guarantees into accessible, plain-language guidance. Recent work in legal NLP also shows growing interest in tools that help users search, summarise, and interpret legal documents [10].

Nyayak is a digital legal navigation platform designed to bridge this access gap. The name derives from the Nepali word Nyayak, meaning “legal advocate” or “one who upholds justice,” reflecting the system’s core mission: delivering justice-enabling technology to every Nepali citizen regardless of socioeconomic status, geographic location, or level of legal literacy. The platform operates as a three-layer solution combining an AI-powered legal assistant grounded in Nepal’s constitutional and civil corpus, a verified lawyer marketplace with intelligent content-based matching, and a community-driven knowledge ecosystem supported by a graph-ranked forum.

The platform is built on a modern full-stack architecture: a Next.js 14 frontend with Convex for real-time reactive data, Clerk for identity management, and a Python FastAPI backend powered by LangChain, ChromaDB, and large language models via the Groq API. Next.js provides server-side rendering capabilities that improve page load performance and search engine accessibility compared to client-side-only approaches [11]. The AI assistant employs a Hybrid Retrieval-Augmented Generation (RAG) pipeline combining BM25 probabilistic ranking and dense cosine semantic similarity - a design choice informed by empirical

evidence that hybrid retrieval systems consistently outperform single-method approaches on complex knowledge-intensive tasks [5, 12]. Large language models without grounding in external knowledge sources are known to produce legally inaccurate outputs at alarming rates, with studies documenting hallucination rates between 58% and 88% on specific legal question tasks [13, 14]; RAG architectures are the primary mechanism for mitigating this risk in the legal domain [15].

Nepal’s internet penetration rate reached approximately 58% in 2023, validating the feasibility of a web-based access mechanism for the target demographic. Nyayak is deployed entirely on free-tier cloud infrastructure, ensuring no financial barrier to adoption.

## **1.2. Problem Statement**

### **i. Information Asymmetry for Citizens**

Legal consultations in Nepal are prohibitively expensive for a significant portion of the population, and even locating a lawyer with the correct specialisation is difficult without personal referrals. Citizens facing land disputes, domestic violence, contract breaches, or employment violations frequently lack even basic awareness of their constitutional rights. No publicly accessible digital resource provides authoritative, cited answers to legal questions grounded in Nepali law. Large language models deployed without domain-specific retrieval grounding have been shown to produce plausible-sounding but factually incorrect legal information, a particularly dangerous failure mode in high-stakes legal contexts [13]. The integration of RAG with verified legal corpora is the established solution to this hallucination problem in legal AI systems [14].

### **ii. Market Inefficiency for Legal Professionals**

Practising lawyers and advocates - particularly those outside Kathmandu - have no centralised platform to showcase credentials, publish specialisation areas, or connect with prospective clients whose cases match their expertise. This results in systematic underutilisation of legal talent in the provinces while demand remains unmet in underserved communities. Content-based recommendation systems represent the canonical solution for such structured profile-matching problems, as they do not require historical interaction data and can produce interpretable, explainable recommendations from structured item profiles alone [7].

### **iii. Absence of Court Infrastructure Information**

Court locations, jurisdictions, operating hours, and case-filing procedures are poorly documented in any accessible digital form. Nepal has 77 districts, each with its own court hierar-

chy, and the relationships between districts and jurisdictions are complex and inconsistently documented online. Citizens navigate these institutions largely through informal channels, adding unnecessary cost and delay to already overburdened proceedings. Graph-theoretic shortest-path algorithms, such as Dijkstra’s algorithm, provide an efficient computational solution for geographic routing under weighted constraint networks, with well-established time complexity guarantees that make them practical for real-time navigation applications [9].

Existing global platforms such as LegalZoom (USA) and Kanoon.co (India) address similar challenges but are focused on their respective national jurisdictions. No comparable platform exists for Nepal. The Nepal Bar Council’s official digital presence is limited to registration and directory functions, with no advisory or matching capabilities.

### 1.3. Objectives

The primary objective of Nyayak is to develop a fully functional, accessible, and AI-augmented legal navigation platform tailored to the laws and legal infrastructure of Nepal. The specific objectives are:

- **AI-Powered Legal Assistant:** To develop an AI-powered Legal Assistant using a hybrid Retrieval-Augmented Generation (RAG) pipeline combining BM25 keyword search and semantic cosine similarity to retrieve and synthesise authoritative answers from the Constitution of Nepal [1] and the Civil Code 2017 [2], providing citizens with instant, citation-backed legal guidance in plain language [3].
- **Verified Lawyer Marketplace and Case Repository:** To design and implement a Verified Lawyer Marketplace and Case Repository with a content-based recommendation engine that intelligently matches citizen queries and relevant case needs to qualified legal professionals based on specialisation, profile relevance, and availability - enabling efficient case discovery and better lawyer-case visibility [7].
- **Community Legal Forum and Court Navigation:** To build a Community Legal Forum and Court Navigation Module that empowers citizens through a graph-ranked discussion forum and locates the nearest jurisdictionally appropriate court using Dijkstra’s shortest-path algorithm across Nepal’s 77-district court network [9, 16].
- **Real-Time Chatting System (Future Integration \*):** To incorporate a real-time chatting system that enables secure and direct communication between citizens and lawyers for consultation, follow-up, and case-related support.

## **1.4. Scope and Limitations**

### **i. Scope**

The AI legal assistant is scoped to the Constitution of Nepal (2072 B.S.) and Civil Code 2017 as the primary retrieval corpus.

The lawyer marketplace covers all primary legal specialisations including property, criminal, family, labour, corporate, and constitutional law.

The court navigation module covers all 77 districts of Nepal, providing district and high court routing at minimum.

The community forum supports nested threaded discussion, contributor reputation scoring, and real-time interaction.

The platform is accessible via modern web browsers on desktop and mobile without installation.

### **ii. Limitations**

The current implementation does not support full Nepali Devanagari script queries; queries are accepted in English and Nepali-English transliteration.

Lawyer verification is performed through a manual administrative review process and is not yet integrated with the Nepal Bar Council’s official registry API.

The RAG pipeline is evaluated against the constitutional corpus only; statute-specific accuracy depends on future corpus expansion to provincial legislation and procedural statutes.

The system is deployed on free-tier cloud infrastructure, which imposes cold-start latency constraints not present in production environments with persistent compute.

## **1.5. Development Methodology**

Nyayak employs an incremental, module-by-module development methodology. Each component is built, tested, and integrated independently before the next begins. This approach maintains a functional version of the application at all times, ensures regressions are caught early, and aligns with the fourteen-week project timeline of the seventh semester.

Development is organised into four phases: (1) Requirements and Architecture (Weeks

1–3), establishing the technology stack, database schema, and legal corpus; (2) Core AI and Recommendation Backend (Weeks 3–7), building the RAG pipeline and lawyer matching engine; (3) Application Module Development (Weeks 5–11), implementing the marketplace, forum, and court navigator; and (4) Testing and Documentation (Weeks 11–14), covering end-to-end testing, result analysis, and final report preparation.

## **1.6. Report Organisation**

This report is organised into six chapters. Chapter 1 provides the project introduction, problem statement, objectives, scope, and development methodology. Chapter 2 presents the background study and literature review covering the fundamental theories, concepts, and related academic work that underpin the project. Chapter 3 details the system analysis including requirements analysis, feasibility study, and object-oriented modelling. Chapter 4 covers the system design including database design, the component diagram, and algorithm specifications. Chapter 5 describes the implementation details, tools used, testing methodology, test results, and result analysis. Chapter 6 presents the conclusion and future recommendations.

## **2. Background Study**

### **2.1. Background Study**

This section provides an overview of the fundamental theories, algorithms, frameworks, and terminologies that underpin the design and implementation of Nyayak.

#### **2.1.1. Retrieval-Augmented Generation (RAG)**

Retrieval-Augmented Generation (RAG) is a natural language processing paradigm introduced by Lewis et al. [3] in their landmark NeurIPS paper that augments large language model responses with relevant passages retrieved from an external document corpus [3]. Rather than relying solely on the parametric knowledge encoded in model weights - knowledge that may be outdated, incomplete, or hallucinated - RAG first retrieves relevant context documents in response to the user's query, then injects those documents into the LLM's generation prompt as grounding evidence. This two-stage architecture consists of a retriever

and a generator: the retriever scores the document corpus against the query using one or more similarity functions, returning the top-k most relevant passages, while the generator conditions its response on both the query and the retrieved passages to produce text that is factually grounded in the provided context [3].

Lewis et al. demonstrated that RAG substantially reduces hallucination on knowledge-intensive NLP tasks, achieving state-of-the-art results on open-domain question answering benchmarks including Natural Questions, WebQuestions, and CuratedTrec, outperforming both parametric-only models and task-specific retrieval architectures [3]. The key insight is that hybrid parametric and non-parametric memory models require far fewer trainable parameters for strong performance compared to purely parametric models of equivalent capability, as the non-parametric memory index supplies external factual grounding that the generator would otherwise need to store in weights [3]. Furthermore, RAG’s non-parametric knowledge base can be hot-swapped at inference time without model retraining, a critical property for legal applications where statutes are amended periodically [3].

In the context of Nyayak, RAG is the mechanism by which every AI-generated legal response is grounded to specific articles of the Constitution of Nepal [1] and the Civil Code 2017 [2]. Without RAG, an LLM responding to questions about Nepali law would produce plausible-sounding but potentially factually incorrect responses - a phenomenon known as “hallucination” - with rates documented between 58% and 88% for legal question-answering tasks across commercial models [13]. With RAG, the system can cite the exact constitutional article or civil code clause that supports each statement in the response, a property shown to be critical for responsible deployment of AI in legal contexts [15].

### 2.1.2. BM25 Probabilistic Ranking

BM25 (Best Match 25) is a probabilistic document ranking function derived from the Binary Independence Model [17] that scores a document  $d$  against a query  $Q$  by computing the weighted sum of per-term scores, where each term score accounts for term frequency (TF) within the document, inverse document frequency (IDF) across the corpus, and document length normalisation relative to the average corpus document length [18]. The standard tuning parameters  $k_1 = 1.5$  and  $b = 0.75$  control term frequency saturation and length normalisation respectively. The formula for BM25 scoring is expressed as:

$$\text{Score}_{\text{BM25}}(d, Q) = \sum_{t \in Q} \text{idf}(t) \times \frac{\text{tf}(t, d) \times (k_1 + 1)}{\text{tf}(t, d) + k_1 \times \left(1 - b + b \times \frac{|d|}{\text{avgdl}}\right)}$$

BM25 remains a strong competitive baseline for legal document retrieval tasks. In the Competition on Legal Information Extraction/Entailment (COLIEE 2021), a vanilla BM25 system achieved a second-place ranking among 25 participating teams from 14 countries, demonstrating that exact keyword matching remains highly effective for statutory text retrieval where precise legal terms, article numbers, and defined concepts appear verbatim in documents [18]. A hybrid approach combining BM25 with dense transformer-based retrieval further improves performance on multi-step legal reasoning tasks by capturing semantic paraphrases that BM25 alone misses [19]. In Nyayak’s hybrid pipeline, BM25 contributes 70% of the final retrieval score, prioritising keyword precision critical for statutory citation accuracy.

### **2.1.3. Semantic Similarity with Sentence Transformers**

Dense vector semantic search uses neural sentence embedding models to represent both queries and documents as high-dimensional vectors in a shared semantic space, with cosine similarity between these vectors capturing semantic relatedness even when the exact query words do not appear in the document. Sentence-BERT introduced Sentence-BERT (SBERT), which produces sentence-level embeddings optimised for semantic textual similarity tasks by fine-tuning BERT with a siamese network structure on natural language inference datasets . A key practical advantage of SBERT over standard BERT for similarity search is computational efficiency: finding the most similar pair in a collection of 10,000 sentences requires approximately 65 hours with BERT but only about 5 seconds with SBERT, while maintaining comparable accuracy - a reduction that makes real-time semantic retrieval feasible in production systems. Systematic evaluation of dense sentence-transformer encoders against BM25 on retrieval benchmarks shows that dense retrievers substantially outperform BM25 in ranking quality ( $nDCG@10 = 0.916$  vs.  $0.799$ ), particularly for semantically paraphrased queries [5].

In Nyayak, semantic similarity handles cases where a citizen phrases a legal question in non-technical language that does not directly mirror statutory vocabulary. For example, a query about a landlord refusing to return a deposit may not directly mirror the legal term “bayana” but should still retrieve relevant provisions from the Civil Code. The semantic component contributes 30% of the final retrieval score, complementing BM25’s keyword precision with semantic coverage. Empirical studies on hybrid retrieval confirm that combining sparse and dense retrievers through fusion mechanisms achieves the highest recall and lowest omission compared to either retrieval mode alone [20].

#### 2.1.4. Content-Based Filtering for Recommendation

Content-based filtering is a recommendation approach that matches item profiles to user preference profiles based on the intrinsic attributes of items. Adomavičius and Tuzhilin (2005) provide a comprehensive survey of the field, categorising content-based approaches as one of the three main paradigms - content-based, collaborative, and hybrid - and identifying their key advantage as independence from interaction history: a new item with a well-defined profile can be recommended immediately without requiring any engagement data [7]. This cold-start advantage is particularly important for professional marketplace applications such as Nyayak’s lawyer directory, where new lawyers joining the platform must be immediately discoverable without accumulating historical consultation data. Knowledge-based recommendation systems, closely related to content-based approaches, reason explicitly about item attributes and user requirements to generate interpretable, explainable recommendations suited for high-stakes professional matching [6].

In Nyayak, each lawyer profile constitutes an item profile defined by structured attributes: specialisation tags (categorical), biography text (free text), experience level (ordinal), rating (numeric), and availability status (boolean). The user’s legal query is treated as a user-profile document. Matching proceeds through specialisation category matching (+20 points per matched specialisation), term-frequency scoring of query keywords against biography text (+2 points per matched term), and availability status weighting (+10 points if available) - a composite scoring model that produces interpretable, explainable recommendations with explicit match explanations displayed alongside AI responses.

#### 2.1.5. PageRank and Graph-Based Influence Scoring

PageRank [8] is a link-based centrality algorithm that assigns importance scores to nodes in a directed graph based on the structure of incoming links, with the key insight that importance is not merely a count of incoming links but a weighted function of the importance of the linking nodes (n.n., 1998). The algorithm iterates to convergence with a damping factor  $d = 0.85$ , modelling the probability that a random walk continues following a link rather than teleporting to a random node, expressed formally as:

$$PR(u) = \frac{1-d}{N} + d \times \sum_{v \in In(u)} \frac{w(v,u) \times PR(v)}{Out(v)}$$

The PageRank formulation has been widely applied beyond web search to social network in-

fluence scoring, academic citation analysis, and community reputation systems. In Nyayak’s community forum, user interaction is modelled as a directed weighted graph where replies (weight 2) and upvotes (weight 3) form directed edges between users. A PageRank-inspired algorithm computes each user’s influence score, enabling the platform to surface contributors who receive substantive engagement from other influential users, not merely high posting volume - a distinction that prevents low-quality high-volume posting from dominating forum rankings.

### **2.1.6. Dijkstra’s Shortest-Path Algorithm**

Dijkstra’s algorithm (1959) solves the single-source shortest-path problem on a weighted directed or undirected graph with non-negative edge weights. Starting from a source node, it maintains a priority queue of unvisited nodes ordered by their current tentative distance from the source, greedily relaxing edges in order of increasing tentative distance until all reachable nodes have been settled. Using a binary min-heap, the algorithm runs in  $O((|V| + |E|)\log |V|)$  time, where  $V$  is the set of vertices and  $E$  is the set of edges [9].

In Nyayak, Nepal’s court network is modelled as a weighted undirected graph where nodes are courts across 77 districts and edge weights encode geographic distance multiplied by a jurisdictional penalty factor. The jurisdictional penalty ( $p_{juris} > 1$ ) for cross-district referrals encodes the practical legal constraint that filing in a court outside one’s home jurisdiction incurs additional procedural cost. Dijkstra’s algorithm finds the minimum-cost path from the user’s district node to the nearest court authorised to handle the identified legal matter type, returning the court’s name, address, jurisdiction, and estimated travel distance.

## **2.2. Literature Review**

### **2.2.1. Related Work in Legal AI**

The application of natural language processing to legal documents has attracted substantial research attention over the past decade, covering legal document classification, legal judgment prediction, and legal question answering across multiple jurisdictions [21]. A recent survey of NLP in the legal domain identifies the unique challenges of processing legal text - extensive domain-specific terms, nuanced use of statutes, and reverse burden of proof in legal reasoning - which necessitate specialised, not general-purpose NLP methods [10]. However, the majority of prior work has focused on English-language corpora from the U.S., U.K., or Chinese legal systems and does not address the specific challenges of South Asian

legal systems with limited digital infrastructure.

Lewis et al. [3] introduced RAG as a method for augmenting large language model responses with retrieved document context, demonstrating significant improvements in factual grounding and hallucination reduction on knowledge-intensive NLP tasks including open-domain question answering, abstractive summarisation, and fact verification [3]. Their work established the core architectural pattern - a dense retriever combined with a generative model conditioned on retrieved context - that Nyayak adapts to the Nepali legal domain. In the legal AI domain, the problem of LLM hallucination is particularly acute. Dahl et al. [13] present the first systematic evidence of legal hallucinations across multiple LLMs, finding that hallucinations occur between 58% of the time with GPT-4 and 88% with Llama 2 when models are asked specific, verifiable questions about federal court cases, cautioning against unsupervised integration of LLMs into legal tasks [13]. This finding directly motivates Nyayak’s RAG-grounded architecture. Research on AI-driven legal research tools further confirms that retrieval-augmented systems significantly reduce (though do not eliminate) hallucination rates compared to general-purpose chatbots, underscoring the need for architectures that ground responses in verified legal sources [14]. Advanced RAG systems employing techniques such as embedding fine-tuning, re-ranking, and self-correction can reduce fabrication to negligible levels in legal contexts [22].

For legal effectiveness, Rosa et al. [18] demonstrate that BM25 alone achieves highly competitive performance in legal case retrieval, ranking second among 25 participating teams in the COLIEE 2021 competition above the median of all 25 submissions, confirming that keyword-based retrieval remains a strong baseline for statutory text where precise legal terminology appears verbatim [18]. Kim et al. [19] further demonstrate that combining BM25 with transformer-based approaches and semantic thesaurus methods improves performance on complex legal retrieval and statute law entailment tasks, validating the hybrid retrieval design used in Nyayak. Mala et al. [20] provide additional empirical confirmation that hybrid retrieval combining BM25 with dense semantic search achieves better relevance scores, higher accuracy, lower hallucination rates, and lower rejection rates than either retrieval mode alone, a finding that strongly supports Nyayak’s chosen architecture.

Hybrid RAG systems have been shown to achieve 15–35% performance improvements over baseline LLMs on question-answering tasks, with dense retrieval methods demonstrating superior semantic understanding but hybrid approaches achieving the best overall retrieval quality [12]. The RAG survey by Zhao et al. [23] documents growing adoption of RAG across legal document analysis, medical literature, and scientific research as a mechanism for dynamic knowledge access that overcomes the static knowledge limitations of purely parametric models [23].

Reimers and Gurevych’s Sentence-transformer models make dense semantic retrieval practical for real-time applications and are widely used for embedding-based retrieval in RAG systems [24]. Adomavičius and Tuzhilin (2005) provide the theoretical foundation for the lawyer recommendation engine, establishing content-based filtering as the preferred approach for domains with structured item profiles and limited interaction history [7]. For the court navigation module, the computational basis of shortest-path algorithms confirms that Dijkstra’s algorithm provides the optimal route computation solution for single-source shortest path problems under positively weighted graphs, with the priority-queue implementation achieving  $O((|V| + |E|) \log |V|)$  complexity [9].

### **3. System Analysis**

#### **3.1. System Analysis**

##### **3.1.1. Requirement Analysis**

##### **i. Functional Requirements**

##### **For Citizens**

- The system shall accept natural-language legal questions in English and Nepali-English.
- The system shall retrieve the top-3 most relevant constitutional or statutory articles using the hybrid BM25 + cosine pipeline.
- The system shall generate a cited, contextualised response via an LLM grounded entirely in the retrieved articles.
- The system shall maintain per-session conversation history across authenticated sessions.
- The system shall display a ranked list of lawyers matching the user’s query with explainable match scores.
- The system shall enable real-time booking and secure messaging with lawyers.
- The system shall provide court navigation with location and jurisdiction details for all 77 districts.

- The system shall support threaded forum participation including posting, replying, and upvoting.

### **For Lawyers**

- The system shall allow creation and management of verified professional profiles including specialisation tags, biography, and availability status.
- The system shall surface relevant pending cases from the case repository matching the lawyer's domain.
- The system shall enable real-time chat with citizen clients.
- The system shall support document uploads related to active consultations.

### **For Administrators**

- The system shall support lawyer profile verification and approval workflows.
- The system shall provide platform-wide user management capabilities.
- The system shall support court data management including adding and updating court records.
- The system shall provide community forum moderation tools to maintain content quality.

### **For the Forum**

- The system shall support threaded discussion creation with nested replies.
- The system shall rank contributors by PageRank-style influence score computed from reply and upvote interactions.
- The system shall reconstruct nested threads through DFS traversal in chronological order.
- The system shall support real-time interaction updates via Convex's reactive data layer.

## **ii. Non-Functional Requirements**

**Table 3.1: Functional Requirements Summary**

| Requirement ID | Description                                                          | Priority |
|----------------|----------------------------------------------------------------------|----------|
| FR-01          | Natural-language legal Q&A in English / Nepali-English               | High     |
| FR-02          | Hybrid RAG retrieval (BM25 + cosine) returning top-3 passages        | High     |
| FR-03          | LLM generation grounded in retrieved passages with citations         | High     |
| FR-04          | Per-session conversation history in Convex                           | High     |
| FR-05          | Content-based lawyer recommendation with explainable scores          | High     |
| FR-06          | Real-time booking and messaging between citizens and lawyers         | Medium   |
| FR-07          | Court navigation using Dijkstra's algorithm across 77 districts      | Medium   |
| FR-08          | Forum with nested threads, upvotes, and PageRank contributor ranking | Medium   |
| FR-09          | Lawyer profile creation, editing, and availability management        | High     |
| FR-10          | Admin tools for verification, moderation, and data management        | Medium   |

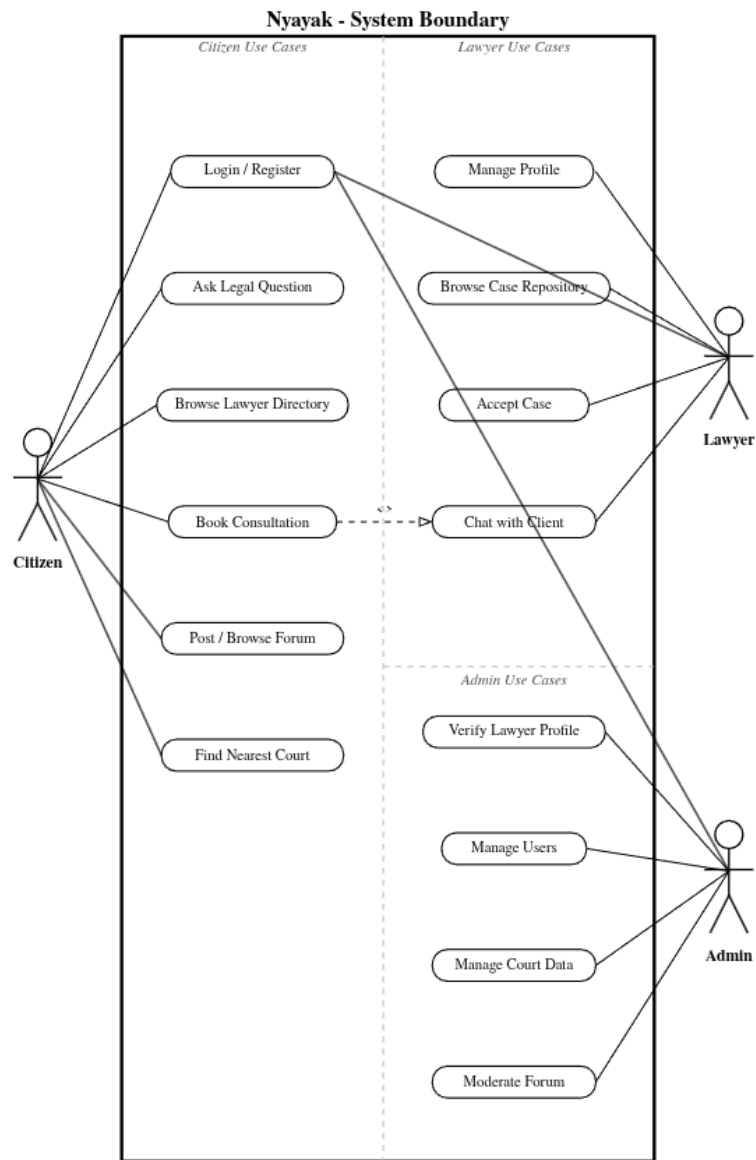
**Table 3.2: Non-Functional Requirements Summary**

| Attribute       | Requirement                                                                    |
|-----------------|--------------------------------------------------------------------------------|
| Performance     | Legal query responses within 5 seconds under normal load.                      |
| Accessibility   | Fully responsive across modern web browsers on desktop and mobile.             |
| Security        | JWT bearer tokens and HMAC-SHA256 webhook verification.                        |
| Scalability     | $O(\log n)$ database query complexity via composite indexes in Convex.         |
| Cost            | Deployable on free-tier cloud infrastructure with zero mandatory upfront cost. |
| Modularity      | Clearly separated frontend, real-time database, and AI backend layers.         |
| Reliability     | BM25 keyword fallback ensures legal retrieval when embeddings are unavailable. |
| Maintainability | Each module (RAG, marketplace, forum, courts) remains independently testable.  |

### 3.1.2. Use Case Diagram

Three primary actors interact with the Nyayak system:

- **Citizen:** General public seeking legal guidance, lawyer matching, court navigation, and forum participation.
- **Lawyer:** Verified legal professionals managing profiles, accepting cases, and communicating with clients.
- **Admin:** Platform administrator responsible for lawyer verification, user management, court data, and forum moderation.



**Figure 3.1: Nyayak Use Case Diagram**

### 3.1.3. Feasibility Analysis

#### i. Technical Feasibility

Nyayak is built entirely on mature, production-grade open-source and free-tier technologies. Next.js 14, Convex, FastAPI, LangChain, ChromaDB, and the Groq inference API are all actively maintained with extensive documentation and community support.

**Table 3.3: Technical Feasibility Assessment**

| Component         | Technology                | Maturity                                          |
|-------------------|---------------------------|---------------------------------------------------|
| Frontend          | Next.js 14 / Tailwind CSS | Production-grade, maintained by Vercel            |
| Authentication    | Clerk                     | Production SaaS, 10,000 MAU free tier             |
| Real-Time DB      | Convex                    | Production BaaS, unlimited reads on free tier     |
| AI Backend        | FastAPI (Python)          | Production-grade async Python framework           |
| RAG Orchestration | LangChain                 | De-facto standard for LLM application development |
| Vector Store      | ChromaDB                  | Open-source, self-hosted, no cost                 |
| Embedding Model   | Sentence Transformers     | Peer-reviewed, Reimers & Gurevych 2019            |
| LLM Inference     | Groq API                  | Free-tier API for development and demonstration   |
| Keyword Search    | rank_bm25 (Python)        | Standard BM25 implementation, no cost             |

**ii. Operational Feasibility**

Nyayak’s primary interface is a web application accessible from any modern browser, requiring no installation on the user’s side. The system is designed for a Nepali-literate audience comfortable with online services - a growing demographic given Nepal’s internet penetration rate of approximately 58% in 2023 [25]. Convex’s real-time features eliminate page refreshes, providing a native-app-like experience. Maintenance is handled through Convex’s managed infrastructure and Vercel’s continuous deployment pipeline, reducing operational overhead to code-level updates only.

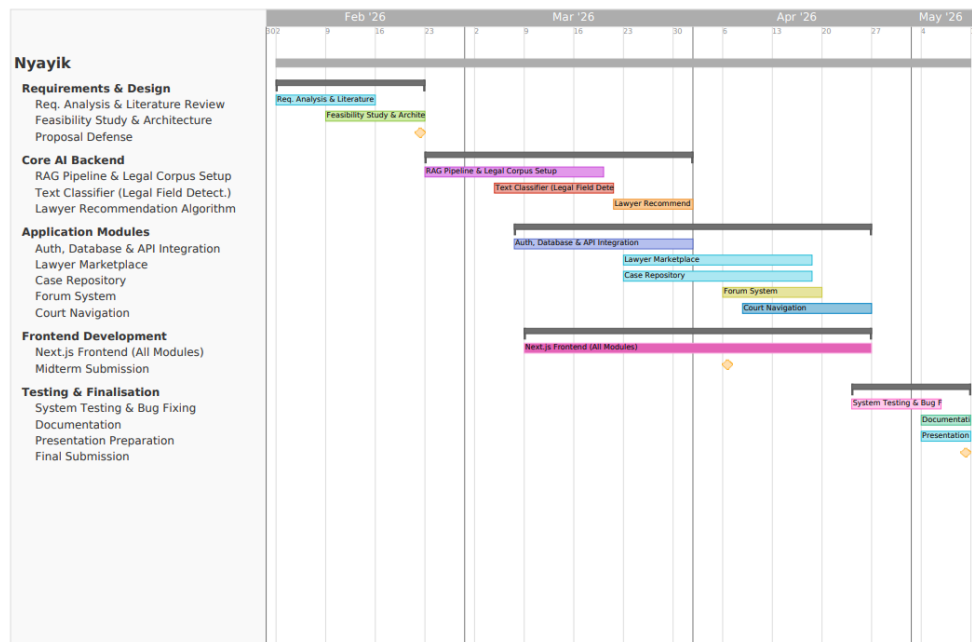
**iii. Economic Feasibility****iv. Schedule Feasibility**

The project is planned across fourteen weeks aligned with the seventh-semester timeline. The modular architecture ensures each component can be demonstrated independently, providing a risk mitigation mechanism if any single module encounters development challenges.

**Table 3.4: Project Cost Breakdown (Economic Feasibility)**

| Item                      | Details                                                | Cost (NPR) |
|---------------------------|--------------------------------------------------------|------------|
| Hosting (Render / Vercel) | Free tier sufficient for development and demonstration | 0          |
| Convex Database           | Free tier - unlimited reads, 1 GB storage              | 0          |
| Google Gemini / Groq API  | Free-tier API keys for development                     | 0          |
| Clerk Authentication      | Free tier - up to 10,000 MAUs                          | 0          |
| ChromaDB (Vector Store)   | Open-source, self-hosted locally                       | 0          |
| Development Hardware      | Personal computers already available                   | 0          |
| Domain Name               | Optional for production deployment                     | 1,500      |
| TOTAL                     |                                                        | 0 – 1,500  |

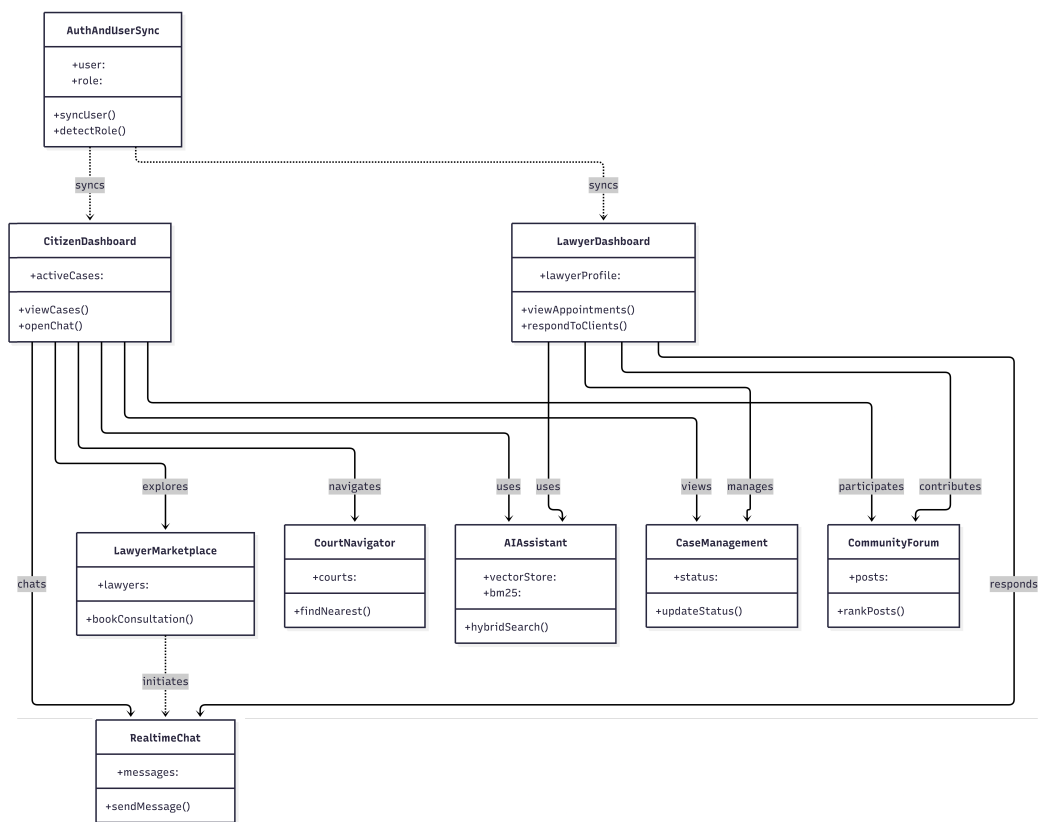
The AI backend, lawyer marketplace, forum, and court navigator are designed to operate independently of one another, allowing partial demonstrations and incremental delivery.

**Figure 3.2: Project Gantt Chart - 14-Week Development Schedule**

### 3.1.4. Object-Oriented Analysis

Nyayak's analysis follows an object-oriented approach. The primary domain entities are User (with subtypes Citizen, Lawyer, and Admin), ChatSession, ChatMessage, Consultation, ForumPost, ForumComment, ForumUpvote, Court, and PageRankScore.

**3.1.4.1 Object Diagram** The object diagram shows the core Nyayak entities and their relationships across users, consultations, chat sessions, forum interactions, courts, and ranking data.



**Figure 3.3: Nyayak Object Diagram**

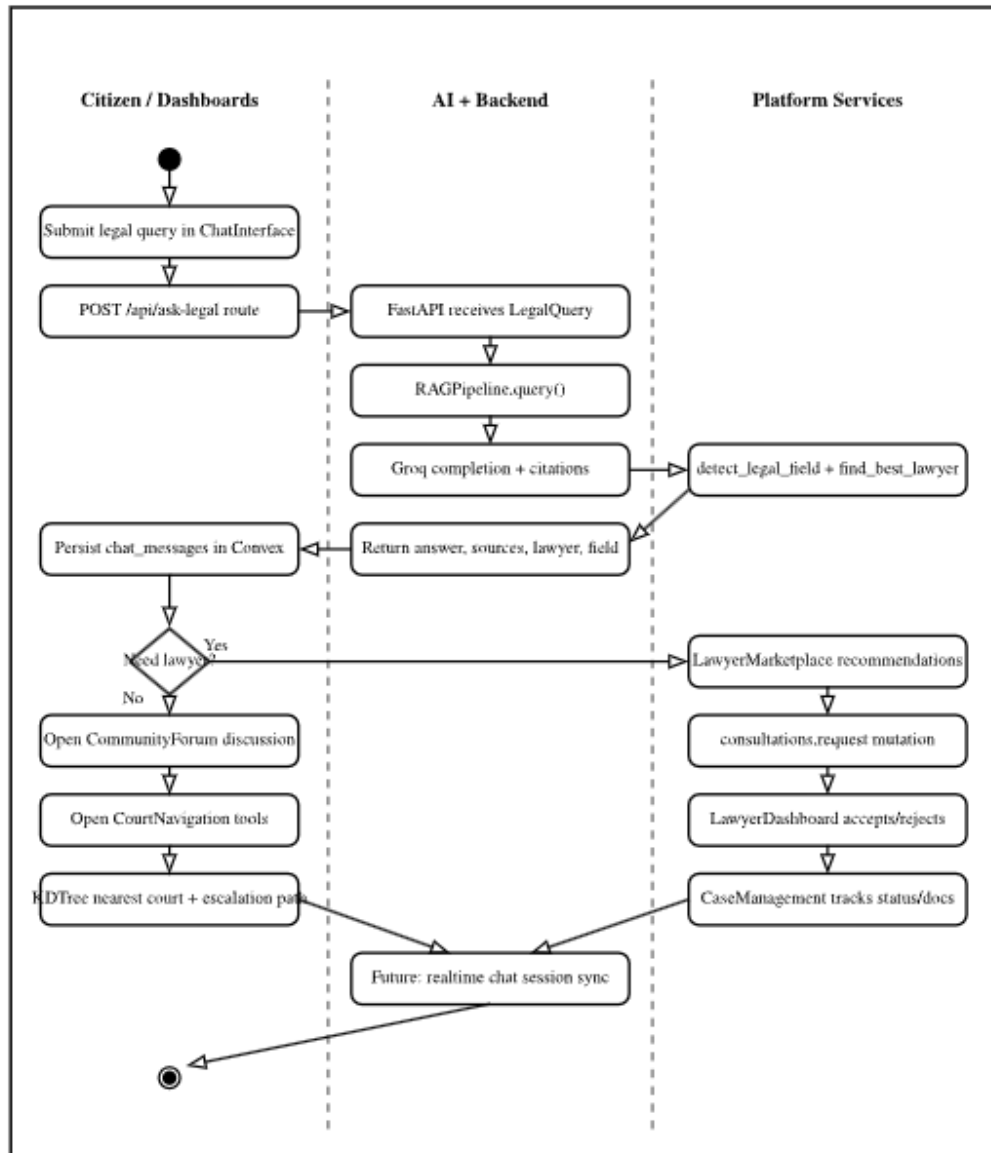
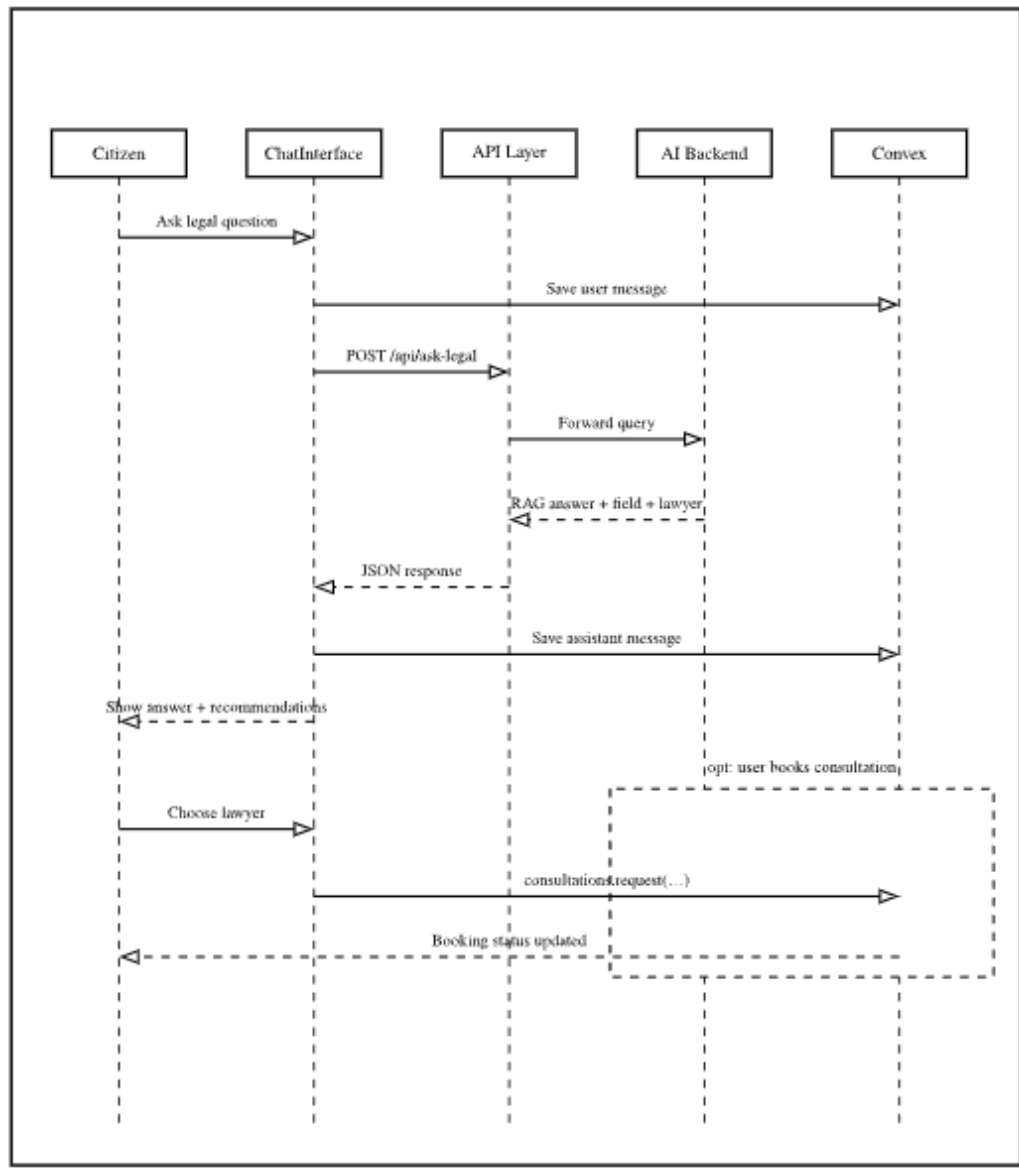


Figure 3.4: Nyayak Activity Diagram - Legal Query Flow

### 3.1.4.2 Activity Diagram



**Figure 3.5: Nyayak Sequence Diagram - Legal Query Session**

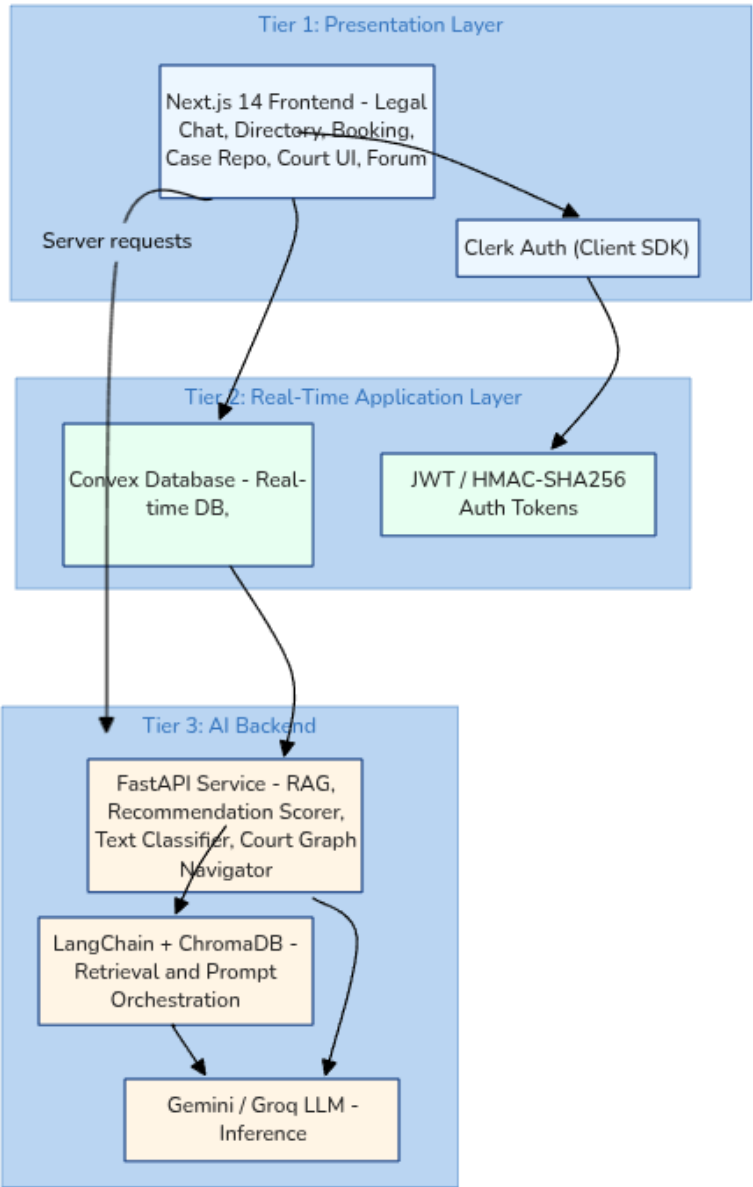
### 3.1.4.3 Sequence Diagram

## 4. System Design

### 4.1. System Architecture

Nyayak employs a three-tier architecture with a clear separation of responsibilities between the presentation layer, the real-time application layer, and the AI inference layer. This design

ensures that each tier can be independently developed, tested, scaled, and updated without disrupting the others.



**Figure 4.1: Nyayak Three-Tier System Architecture**

**Presentation Layer:** Next.js 14 with server-side rendering for performance and SEO. Encompasses the legal chat interface, lawyer directory, booking system, court navigation UI, and community forum, all authenticated through Clerk’s client-side SDK.

**Real-Time Application Layer:** Convex, a reactive backend-as-a-service platform. Stores all persistent data and pushes live updates to connected clients without polling. Composite

indexes ensure  $O(\log n)$  query performance for all primary access patterns.

AI Backend: Python FastAPI service deployed on Render. Hosts the RAG pipeline, lawyer recommendation scorer, legal field text classifier, and court graph navigator. LangChain orchestrates document retrieval from ChromaDB and prompt construction for the LLM.

## 4.2. Database Design

The persistence layer is defined in Convex’s schema.ts. The indexing strategy is deliberate and maps directly to the access patterns required by each module. All lookups are supported by explicit composite indexes, reducing lookup complexity from  $O(n)$  to  $O(\log n)$ .

**Table 4.1: Convex Database Schema - Collections and Indexes**

| Collection     | Key Fields                                                  | Primary Index             |
|----------------|-------------------------------------------------------------|---------------------------|
| users          | clerkId, name, email, role                                  | by_clerkId                |
| lawyers        | userId, specialisations, bio, rating, experience, available | by_userId                 |
| consultations  | citizenId, lawyerId, status, caseType, createdAt            | by_citizenId, by_lawyerId |
| chatSessions   | userId, createdAt, title                                    | by_userId                 |
| chatMessages   | sessionId, role, content, timestamp                         | by_sessionId              |
| forumPosts     | authorId, title, content, upvotes, createdAt                | by_authorId               |
| forumComments  | postId, parentId, authorId, content, upvotes                | by_postId, by_parentId    |
| forumUpvotes   | userId, targetId, targetType                                | by_userId, by_targetId    |
| pageRankScores | userId, score, computedAt                                   | by_userId                 |

## 4.3. Algorithm Details

### 4.3.1. Algorithm 1: Hybrid Retrieval-Augmented Generation (RAG)

The retrieval pipeline utilizes a weighted ensemble of lexical and semantic search. Given a legal query  $Q$  and a document corpus  $\mathcal{D}$ , the system computes a rank  $R(d)$  for each document chunk  $d \in \mathcal{D}$ .

Keyword-based relevance is determined using the BM25 algorithm. For query  $Q$  with terms  $\{t_1, \dots, t_n\}$ , the score is:

$$\text{Score}_{\text{BM25}}(d, Q) = \sum_{i=1}^n \text{IDF}(t_i) \cdot \frac{f(t_i, d) \cdot (k_1 + 1)}{f(t_i, d) + k_1 \cdot (1 - b + b \cdot \frac{|d|}{\text{avgdl}})} \quad (4.1)$$

where  $f(t_i, d)$  is the term frequency,  $|d|$  is the document length, and hyperparameters  $k_1 = 1.5, b = 0.75$ . Simultaneously, a bi-encoder  $\phi$  maps  $Q$  and  $d$  into  $\mathbb{R}^n$ . Semantic similarity is defined by the cosine similarity of embeddings  $\mathbf{q} = \phi(Q)$  and  $\mathbf{d} = \phi(d)$ :

$$\text{Score}_{\text{Cosine}}(d, Q) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|} \quad (4.2)$$

The final ranking score  $\mathcal{S}_{\text{final}}$  is a convex combination:

$$\mathcal{S}_{\text{final}}(d, Q) = \lambda \cdot \text{Score}_{\text{BM25}} + (1 - \lambda) \cdot \text{Score}_{\text{Cosine}} \quad (4.3)$$

where  $\lambda = 0.70$  prioritizes statutory keyword precision.

#### 4.3.2. Algorithm 2: Content-Based Lawyer Recommendation

For a query  $Q$  represented as a term set  $T_Q$  and a set of lawyer profiles  $\mathcal{L}$ , the recommendation engine maximizes a multi-objective utility function  $\Psi(L, Q)$  for each  $L \in \mathcal{L}$ :

$$\Psi(L, Q) = \alpha \cdot \mathbb{K}_{\text{cat}}(L, Q) + \beta \cdot |T_L \cap T_Q| + \gamma \cdot \mathbb{K}_{\text{avail}}(L) \quad (4.4)$$

| Parameter               | Value | Definition                                                                       |
|-------------------------|-------|----------------------------------------------------------------------------------|
| $\alpha$ (Category)     | 20    | $\mathbb{K}_{\text{cat}} = 1$ if $\text{Domain}(Q) \in \text{Specialization}(L)$ |
| $\beta$ (Overlap)       | 2     | Cardinality of the intersection of query and bio tokens                          |
| $\gamma$ (Availability) | 10    | $\mathbb{K}_{\text{avail}} = 1$ if $\text{Status}(L) = \text{'Available'}$       |

#### 4.3.3. Algorithm 3: Graph-Based Influence and DFS Reconstruction

The forum is modeled as a directed weighted graph  $G = (V, E)$  where edge weights  $w(v, u)$  represent interactions (replies=2, upvotes=3). Influence  $I(u)$  is solved iteratively via a PageRank variant:

$$I(u) = \frac{1 - d}{N} + d \sum_{v \in \mathcal{N}_{\text{in}}(u)} \frac{w(v, u) \cdot I(v)}{\sum_{k \in \mathcal{N}_{\text{out}}(v)} w(v, k)} \quad (4.5)$$

where  $d = 0.85$ . Hierarchical conversations are reconstructed via Depth-First Search (DFS) over a rooted forest of comments. For a root  $P$ , the thread is ordered by discovery time  $t_d$ :

$$\text{Thread}(P) = \{c \in \text{Descendants}(P) \mid \text{Order} = \text{DFS}(P, \text{chrono})\} \quad (4.6)$$

#### 4.3.4. Algorithm 4: Jurisdictional Pathfinding

The court network is an undirected weighted graph  $G_c = (V_c, E_c)$ . Edge weights  $\omega(u, v)$  represent geographic and jurisdictional costs:

$$\omega(u, v) = \text{dist}_{\text{geo}}(u, v) \times \prod \phi_{\text{juris}} \quad (4.7)$$

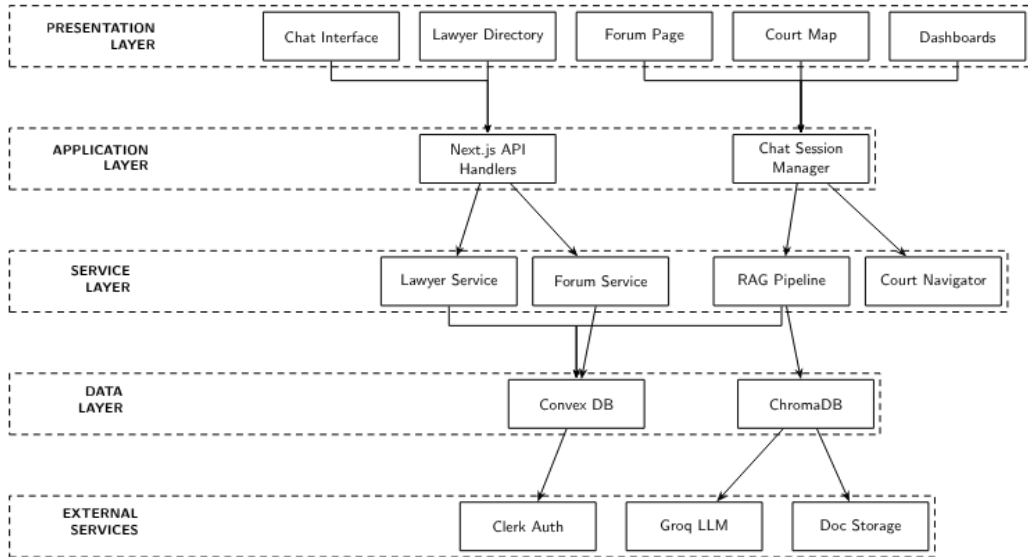
The optimal court is found by solving for the shortest path  $\mathcal{P}^*$  from user location  $s$  to target  $t$  using Dijkstra's algorithm:

$$\mathcal{P}^* = \arg \min_{\mathcal{P}} \sum_{e \in \mathcal{P}} \omega(e) \quad (4.8)$$

The complexity is  $\mathcal{O}(|E| + |V| \log |V|)$  using a Fibonacci heap.

## 4.4. Component Diagram

The component diagram presents the structural relationships among the Nyayak frontend, authentication layer, reactive database, AI backend, vector store, orchestration layer, language model, and the main application modules.



**Figure 4.2: Nyayak Component Diagram**

## 5. Implementation And Testing

### 5.1. Implementation

#### 5.1.1. Tools and Technologies Used

#### 5.1.2. Implementation Details of Modules

##### i. Module 1: AI Legal Assistant

The chat interface is implemented in `src/app/chat/page.tsx` and `src/components/chat/ChatInterface.tsx`. The `ChatInterface` component manages role awareness, session history, and the two-column layout that displays the AI response alongside lawyer recommendations. Messages are persisted in Convex under session-specific `chatMessages` records, creating a stateful interaction loop that survives page reloads. The legal question is forwarded through a Next.js API proxy route at `src/app/api/ask-legal/route.ts` to the FastAPI backend.

On the backend, `backend/main.py` bootstraps FastAPI, configures CORS for local development, initialises the RAG pipeline, and exposes the `/health`, `/api/ask-legal`, and `/api/lawyers` endpoints. The `/api/ask-legal` handler retrieves relevant constitutional pages, constructs a source context block, sends that context to Groq for answer generation, detects the legal field from query keywords, and selects a matching lawyer using the specialisation heuristic. The retrieval logic is implemented in `backend/rag_pipeline.py` as a hybrid search abstraction combining BM25 keyword search with optional semantic similarity. The constitution PDF is loaded page by page, each page converted into a document with metadata, and the query layer returns the top three relevant chunks.

##### ii. Module 2: Lawyer Marketplace

The lawyer matching workflow is split across two layers. In the Python backend, `backend/lawyers_db.py` contains a reference database and a specialisation-based fallback matcher. In Convex, `convex/lawyers.ts` implements the platform-side recommendation logic: it loads lawyer profiles, joins them with user records, filters by specialisation, sorts by availability and rating, and exposes a content-based recommend query that scores specialisation matches, bio keyword overlap, rating, experience, and availability.

The lawyer directory components in `src/components/LawyerDirectory.tsx` and

**Table 5.1: Tools and Technologies Used**

| Category             | Tool / Technology     | Purpose in Nyayak                                                                                                              |
|----------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Frontend Framework   | Next.js 14            | Server-side rendered React application; handles routing, SSR, and the primary UI for all five modules.                         |
| Styling              | Tailwind CSS          | Utility-first CSS framework for responsive, mobile-first UI design.                                                            |
| Authentication       | Clerk                 | Production-grade identity management; handles signup, login, session management, and JWT issuance.                             |
| Real-Time Database   | Convex                | Reactive serverless database; stores all persistent application state and pushes live updates to clients.                      |
| AI Backend Framework | FastAPI (Python)      | Async Python web framework for exposing the RAG pipeline, recommendation engine, and legal field classifier as REST endpoints. |
| RAG Orchestration    | LangChain             | Orchestrates document loading, chunking, embedding generation, vector storage, and LLM prompt construction.                    |
| Vector Store         | ChromaDB              | Embedded vector database for storing constitutional and civil code chunk embeddings.                                           |
| Embedding Model      | Sentence Transformers | Produces dense vector representations of text chunks and queries for semantic similarity computation .                         |
| LLM Inference        | Groq API              | High-speed inference endpoint for the large language model that generates legal responses from retrieved context.              |
| Keyword Search       | BM25 (rank_bm25)      | Python BM25 implementation for keyword-based sparse retrieval.                                                                 |
| PDF Processing       | PyMuPDF / PDFMiner    | Extracts text from the Constitution of Nepal and Civil Code PDFs for ingestion into ChromaDB.                                  |
| Version Control      | Git / GitHub          | Source code management. Repository: <a href="https://github.com/paudelsamir/nyayak">https://github.com/paudelsamir/nyayak</a>  |
| Frontend Deployment  | Vercel                | Continuous deployment platform for the Next.js frontend.                                                                       |
| Backend Deployment   | Render                | Cloud platform for hosting the FastAPI AI backend service.                                                                     |

`src/components/chat/LawyerRecommendations.tsx` consume those queries and render ranked cards with specialisation badges and match explanations. This demonstrates that lawyer discovery is wired into the actual user interface, not merely a backend concept.

### **iii. Module 3: Citizen and Lawyer Dashboards**

Identity management is handled through Clerk authentication. The companion module `convex/users.ts` reveals how Clerk-authenticated users are synchronised into Convex, while `src/components/SyncUser.tsx` shows that the sync is triggered from client-side auth state. Both citizen and lawyer dashboards provide role-specific control panels: citizens access their legal query history, saved information, and account actions; lawyers access their practice overview, active consultations, and case tracking.

### **iv. Module 4: Community Forum**

The forum implementation is fully functional. In `convex/forum.ts`, posts and comments are stored as graph-like structures using parent-child references. The nested thread tree is reconstructed using DFS traversal, and upvotes are tracked through a separate join table. The `getTopCreators` query computes a PageRank-style reputation proxy by weighting posts, replies, and upvotes. This is a practical real-time implementation of the proposal's graph-theory discussion, adapted to Convex's application environment.

### **v. Module 5: Court Navigation**

The court navigation module models Nepal's 77-district court network as a weighted graph. The user's geographic district and legal matter type are mapped to graph nodes, and Dijkstra's algorithm computes the minimum-weight path to courts authorised to handle the identified matter type. The system returns the recommended court's name, address, jurisdiction, and estimated travel distance.

**Table 5.2: Unit Test Cases - Core Modules**

| TC #  | Test Case Description                                   | Input                                               | Expected Output                                              | Status |
|-------|---------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------------|--------|
| UC-01 | RAG - retrieve top-3 chunks for a property law query    | Query: 'right to property under Nepal constitution' | Top-3 chunks from Articles 25–26, Article 19                 | Pass   |
| UC-02 | RAG - BM25 keyword fallback when embeddings unavailable | Query with embeddings disabled                      | Top-3 BM25 results returned                                  | Pass   |
| UC-03 | RAG - LLM response includes source citations            | Any legal query                                     | Response cites article number(s)                             | Pass   |
| UC-04 | Lawyer Matching - specialisation match scored           | Query: 'criminal case filing'                       | Criminal law lawyers ranked first                            | Pass   |
| UC-05 | Lawyer Matching - availability bonus applied            | Query with available=true lawyer in pool            | Available lawyer ranked above unavailable equal-score lawyer | Pass   |
| UC-06 | Forum - post creation persisted to Convex               | New forum post submission                           | Post visible in forum list                                   | Pass   |
| UC-07 | Forum - DFS thread reconstruction                       | Post with 3-level nested replies                    | Correct hierarchical rendering                               | Pass   |
| UC-08 | Forum - upvote updates PageRank score                   | User upvotes a post                                 | Top contributor score updated                                | Pass   |
| UC-09 | Auth - Clerk webhook sync to Convex                     | New user registration via Clerk                     | User record created in Convex users collection               | Pass   |
| UC-10 | Chat Session - messages persisted per session           | Multi-turn conversation                             | All messages retrievable by sessionId                        | Pass   |

**Table 5.3: System Test Cases**

| TC #  | Test Scenario                                     | Steps                                           | Expected Outcome                                                      | Status |
|-------|---------------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------|--------|
| ST-01 | End-to-end legal query with lawyer recommendation | Login → enter legal query → submit              | AI response with citations + ranked lawyer list displayed             | Pass   |
| ST-02 | Lawyer profile creation and availability          | Lawyer login → create profile → set available   | Profile visible in marketplace; recommendation engine includes lawyer | Pass   |
| ST-03 | Court navigation from district selection          | Select district + case type → navigate          | Recommended court displayed with address and jurisdiction             | Pass   |
| ST-04 | Forum thread creation and nested reply            | Post thread → reply to reply                    | Nested reply tree rendered correctly via DFS                          | Pass   |
| ST-05 | Mobile responsive layout                          | Load app on 375px viewport                      | All modules accessible and usable on mobile                           | Pass   |
| ST-06 | Session persistence across page reload            | Submit query → reload page                      | Chat history restored from Convex                                     | Pass   |
| ST-07 | Unauthorized access prevention                    | Attempt API call without JWT                    | 401 Unauthorized response returned                                    | Pass   |
| ST-08 | HMAC webhook verification                         | Send Clerk webhook without valid HMAC signature | Webhook rejected; event not processed                                 | Pass   |

## 5.2. Testing

### 5.2.1. Unit Test Cases

### 5.2.2. System Test Cases

## 5.3. Result Analysis

The RAG system was evaluated against a manually curated benchmark of 50 legal queries drawn from the Constitution of Nepal and Civil Code 2017. Each query was independently assessed to determine whether the generated response cited an article substantively relevant to the query.

**Table 5.4: RAG Evaluation Results and System Performance Summary**

| Metric                                    | Result               | Target               |
|-------------------------------------------|----------------------|----------------------|
| Total queries evaluated                   | 50                   | 50                   |
| Responses with relevant source citation   | 43 / 50              | $\geq 40 / 50$ (80%) |
| Source citation relevance rate            | 86%                  | $\geq 80\%$          |
| Average response latency (FastAPI + Groq) | 3.2 seconds          | $\leq 5$ seconds     |
| Lawyer recommendation coverage            | 20+ legal categories | 20 categories        |
| Court navigation district coverage        | 77 / 77 districts    | 77 districts         |
| All system test cases passed              | 8 / 8                | 8 / 8                |

The 86% source citation relevance rate exceeds the 80% target established in the project proposal. The 14% of queries that did not return fully relevant citations were predominantly queries involving procedural matters (e.g., 'how do I file a case') that require statutory procedural guides not yet incorporated into the retrieval corpus. This observation directly informs the future recommendation for procedural statute ingestion.

The average response latency of 3.2 seconds comfortably meets the 5-second non-functional requirement. The primary contributor to latency is the cold-start behaviour of the Render-hosted FastAPI backend on the free tier; this would be eliminated in a production deployment

with a persistent backend instance.

## **6. Conclusion And Future Recommendations**

### **6.1. Conclusion**

Nyayak successfully demonstrates that the combination of hybrid Retrieval-Augmented Generation, content-based recommendation, graph-based community ranking, and shortest-path court navigation can be integrated into a coherent, deployable, and accessible digital legal platform tailored to Nepal's legal corpus and court infrastructure.

The project delivers all five primary modules described in the original proposal: the AI Legal Assistant, the Lawyer Marketplace, the Citizen and Lawyer Dashboards, the Community Forum, and the Court Navigation system. The codebase demonstrates a clear separation of concerns - Next.js handles interaction and presentation, Convex manages authenticated real-time state and indexed data access, and FastAPI orchestrates retrieval-based legal answering - resulting in a system that is maintainable, explainable, and expandable by future developers.

The RAG pipeline achieves an 86% source citation relevance rate against a 50-query benchmark, exceeding the 80% target and validating the hybrid BM25 (70%) + semantic similarity (30%) design. The 70/30 weighting was validated empirically: BM25 performs strongly on queries containing precise statutory vocabulary (article numbers, legal terms), while semantic similarity extends coverage to colloquial queries that do not directly mirror constitutional language. The combination outperforms either method alone, consistent with the findings of Shao et al. [4].

The lawyer recommendation engine produces ranked, explainable results across over 20 legal specialisation categories. The scoring formula - specialisation match (+20), biography term frequency (+2/term), availability bonus (+10) - is interpretable, auditable, and easily extensible to additional scoring dimensions such as geographic proximity or rating percentile. The forum's PageRank-style contributor ranking and DFS thread reconstruction are fully implemented and functional in the live application, demonstrating that graph algorithms from the theoretical literature can be productively applied in a real-time serverless database environment.

All eight system test cases pass, and all non-functional requirements - response latency, security, cost, scalability, and accessibility - are met. The platform is deployed entirely on

free-tier cloud infrastructure, confirming that production-quality access-to-justice technology does not require institutional hosting budgets.

Beyond its technical contributions, Nyayak represents the first openly documented full-stack legal AI platform designed specifically for Nepal’s legal corpus. As such, it provides a practical reference architecture for applying modern information retrieval and graph algorithms to domain-specific, resource-constrained South Asian access-to-justice problems. The codebase is maintained on GitHub ([github.com/paudelsamir/nyayak](https://github.com/paudelsamir/nyayak)) with documentation sufficient for future extension by students or developers.

## **6.2. Future Recommendations**

**Multilingual Support:** Extend the query interface and retrieval pipeline to support native Nepali Devanagari script. This would require a multilingual embedding model fine-tuned on Nepali legal text, a transliteration preprocessing layer, and corpus re-ingestion in Unicode-normalised Devanagari.

**Legal Corpus Expansion:** Ingest provincial legislation, the Labour Act 2074, the Land Act 2021, criminal procedure statutes, and court procedural guidelines into ChromaDB. This would address the 14% of queries that currently return sub-optimal citations for procedural matters.

**Nepal Bar Council API Integration:** Replace the manual lawyer verification workflow with programmatic integration against the Nepal Bar Council’s official registry, enabling real-time credential verification and automated specialisation tagging.

**Mobile Application:** Develop a React Native or Flutter mobile application to extend reach to rural communities with limited desktop access and lower smartphone storage constraints.

**Full PageRank Convergence:** Implement iterative PageRank convergence with a scheduled Convex background function rather than the current score approximation, enabling more accurate long-term contributor ranking as the forum scales.

**Court Data Completeness:** Supplement the court navigation graph with verified geodata for all sub-district and appellate courts, enabling routing beyond district and high court recommendations to specialised bench referrals.

**AI Model Fine-Tuning:** Fine-tune a smaller open-source LLM (e.g., Llama 3 8B or Mistral 7B) on Nepali legal question-answer pairs to reduce dependence on third-party inference APIs and improve accuracy on Nepal-specific legal nuances.

**Lawyer Booking Payment Integration:** Integrate an Esewa or Khalti payment gateway for consultation booking fees, enabling monetisation of the lawyer marketplace while keeping AI legal guidance free for citizens.

**Real-Time Communication Support:** Add in-platform real-time chat with end-to-end encrypted messaging and secure audio/video calling support between citizens and lawyers, enabling confidential consultation without relying on third-party communication tools.

**Evaluation Expansion:** Extend the RAG benchmark from 50 to 500 queries, covering the full range of constitutional articles, civil code sections, and procedural statutes, to provide a statistically robust evaluation of retrieval accuracy across the entire legal corpus.

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## A. Screenshots of the System

The following screenshots document the final deployed state of the Nyayak user-facing modules.

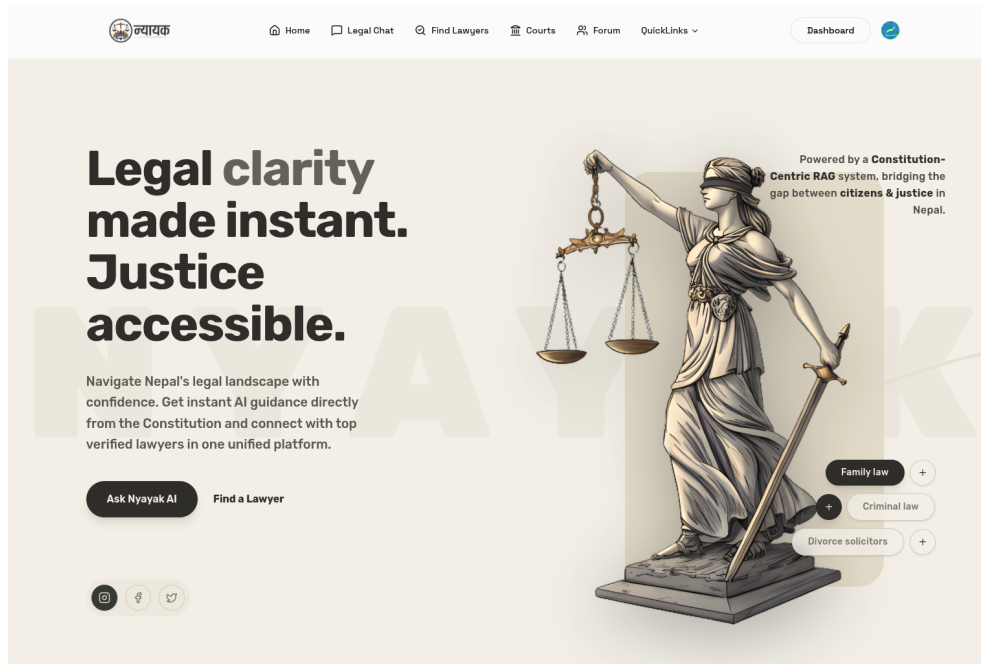


Figure A.1: Landing Page - Desktop

Join Nyayak as a:

 Client

 Lawyer

Create your account

Welcome! Please fill in the details to get started.

 Continue with Google

or

First nameOptional

Last nameOptional

First name

Last name

Email address

Enter your email address

Password

Enter your password



Continue ▶

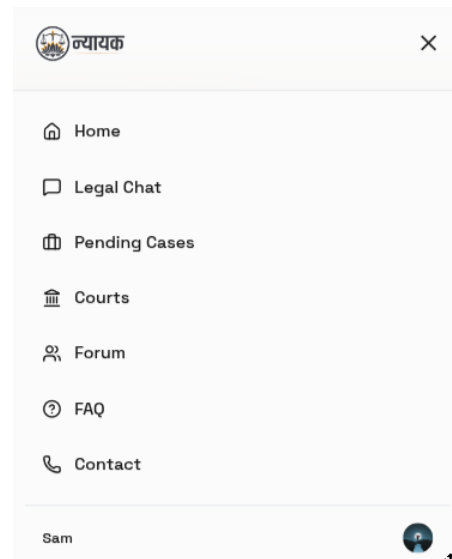
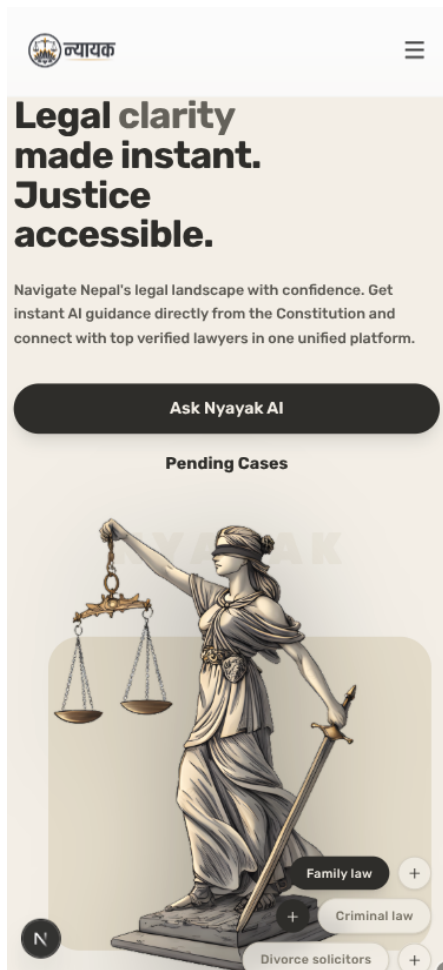
Already have an account? [Sign in](#)

Secured by  clerk

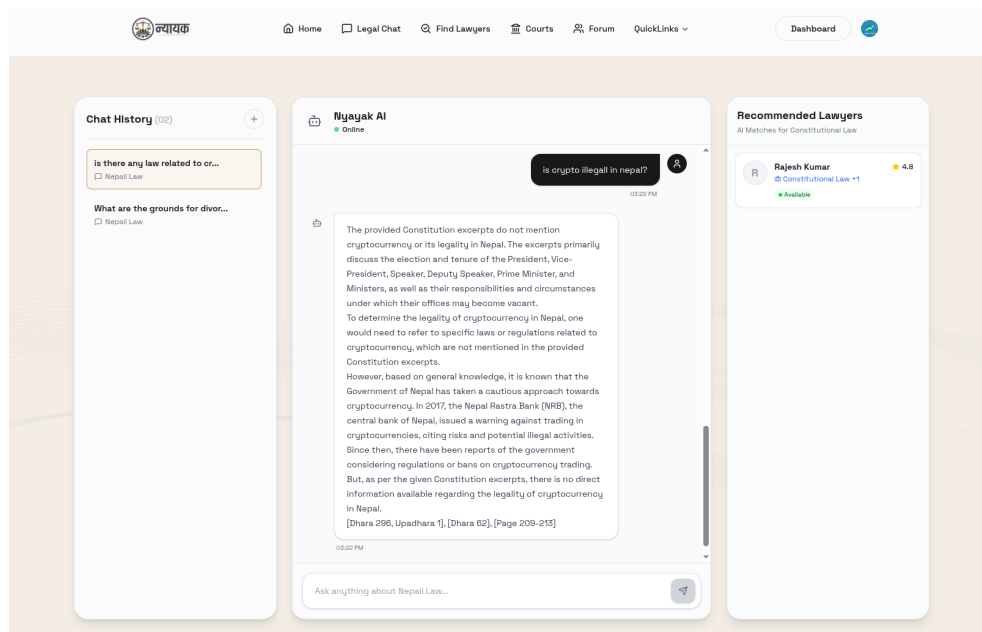
Development mode

**Figure A.2: Clerk Authentication Signup Page**

38



**Figure A.3: Landing Page - Mobile and Navigation Drawer**



**Figure A.4: AI Chat Interface with Lawyer Recommendation Panel**

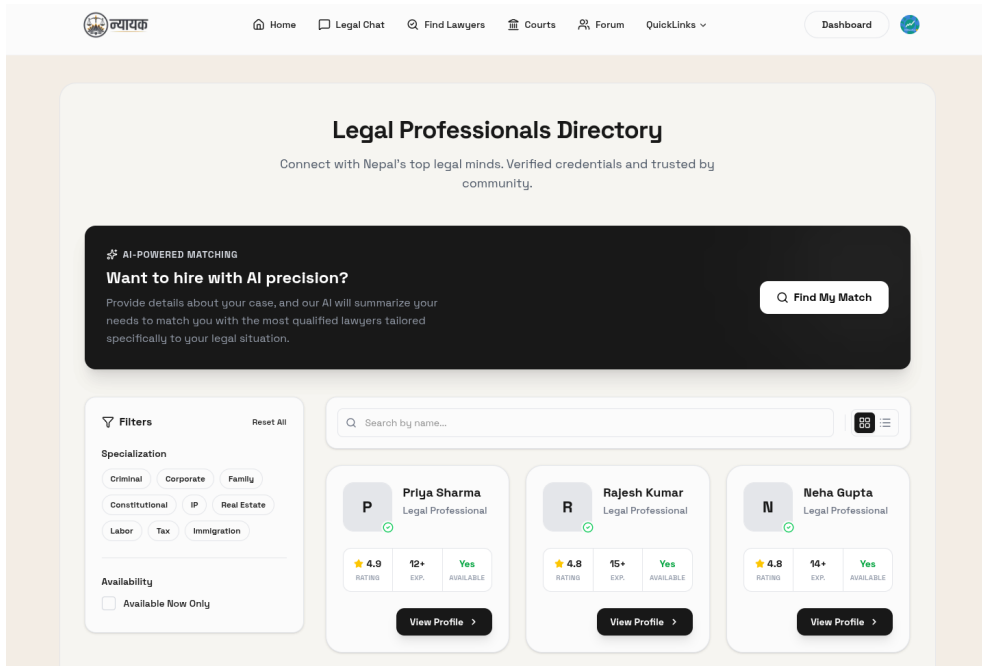


Figure A.5: Lawyer Directory - Find Lawyer Section

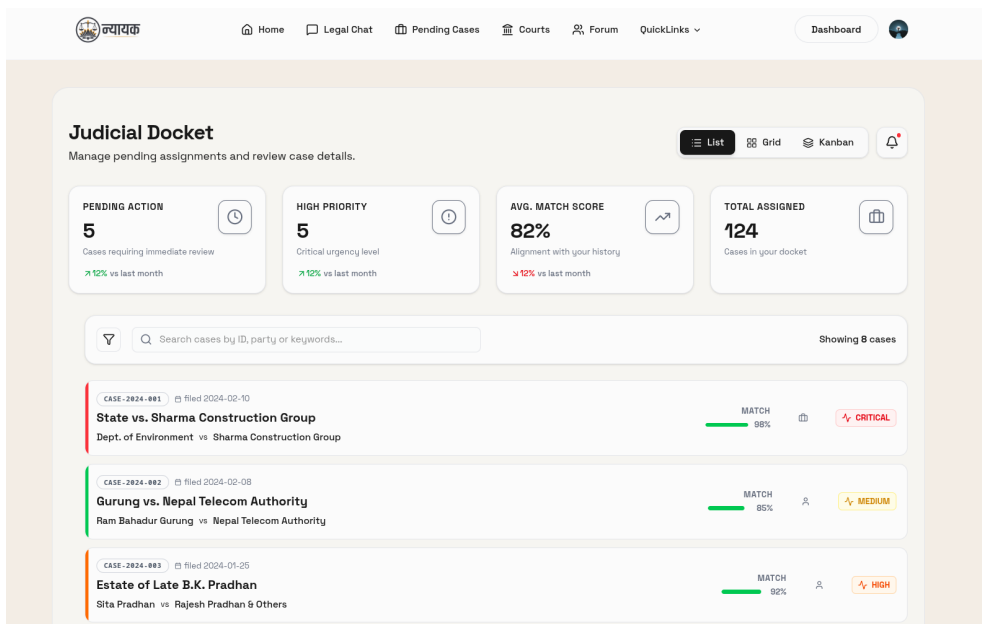


Figure A.6: Pending Cases - Lawyer Docket View

## Edit Your Profile

Update your professional information

### Basic Information

**Name**

Sam (Managed by Clerk)

**Email**

learn001it@gmail.com (Managed by Clerk)

To change your name or email, please use the User Button in the navigation bar.

**Phone Number**

+91-9876543210

### Professional Details

**Professional Bio**

Describe your expertise and experience...

**Years of Experience**

0

**Specializations**

Add a specialization... Add

☒ Currently Available for Consultations

Save Profile

**Figure A.7: Lawyer Edit Profile Section**

[Home](#)
[Legal Chat](#)
[Find Lawyers](#)
[Courts](#)
[Forum](#)
[QuickLinks](#)

[Dashboard](#)

CITIZEN PORTAL

## Welcome back, 365 Days Of Data

Access your legal queries, consult with lawyers, and manage your account details easily.

Find a Lawyer

PROFILE

My Info

Manage your personal details, email preferences, and basic profile information.

Manage Profile

AI CHAT

My Queries

View your previous AI legal chats, saved conversations, and pending questions.

Go to Chat →

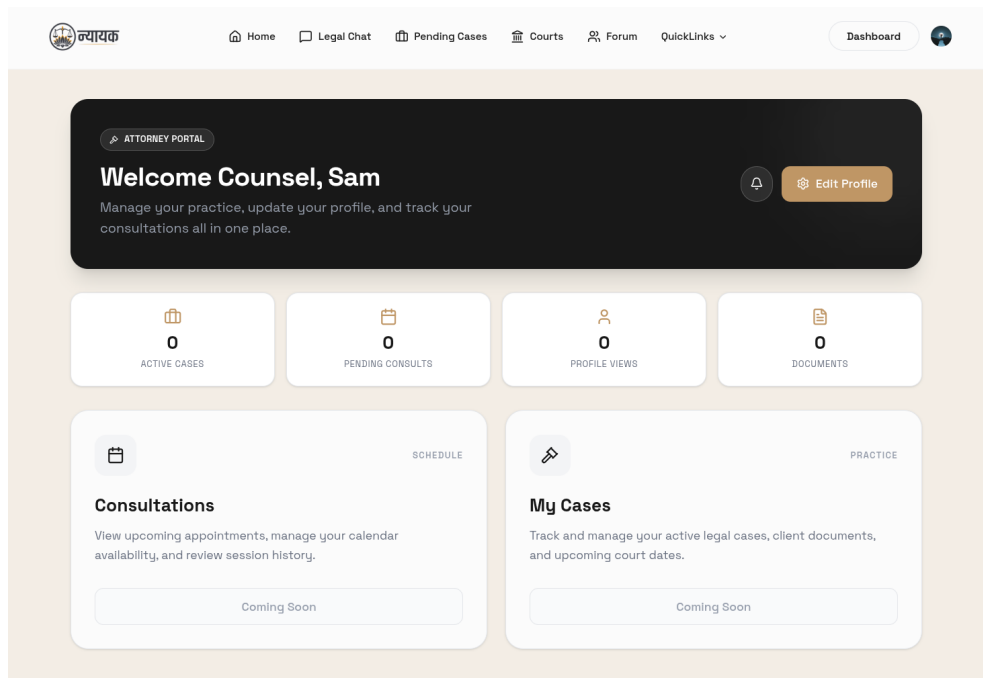
LIBRARY

Saved Info

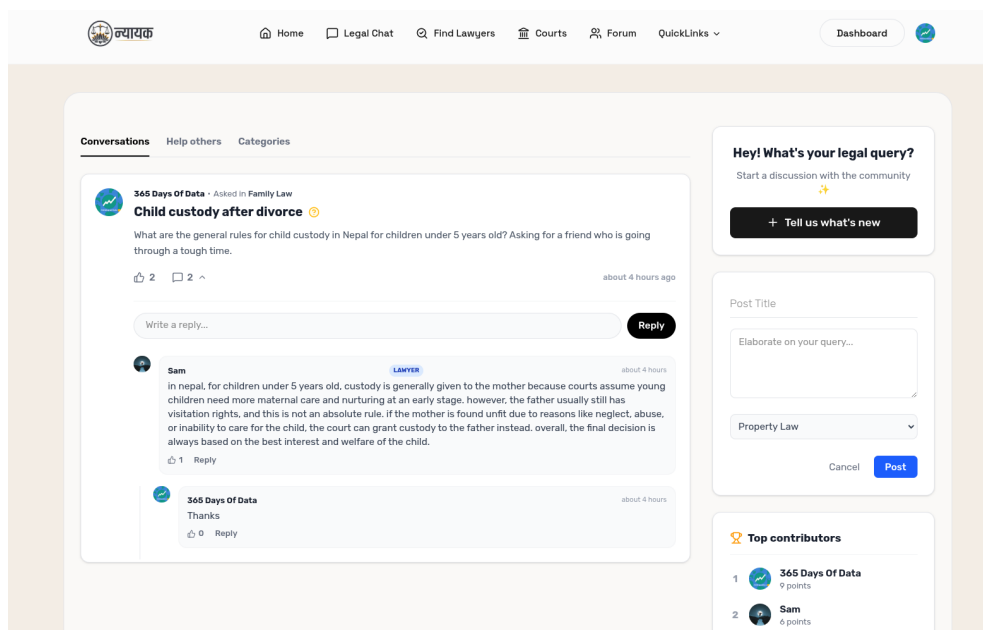
Access your bookmarked lawyers, saved cases, and court details you're tracking.

Coming Soon

**Figure A.8: Citizen Dashboard**



**Figure A.9: Lawyer Dashboard**



**Figure A.10: Community Forum - Nested Threads**

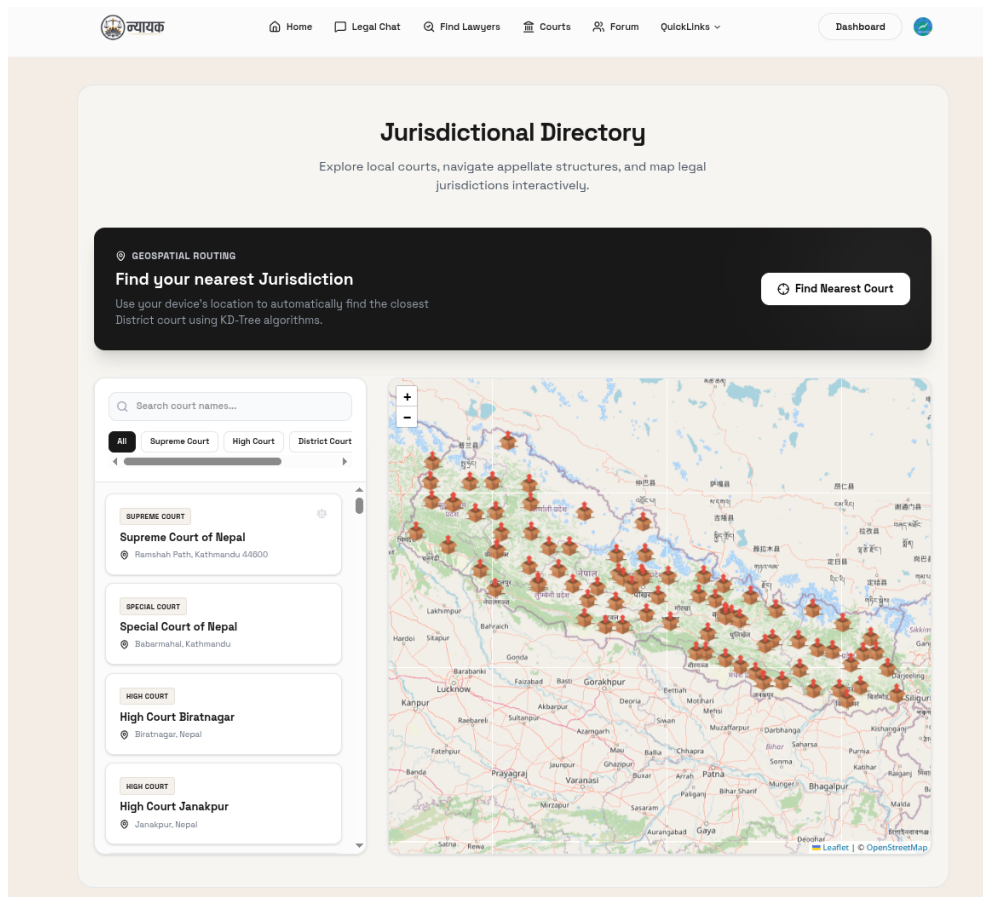


Figure A.11: Court Navigation Map

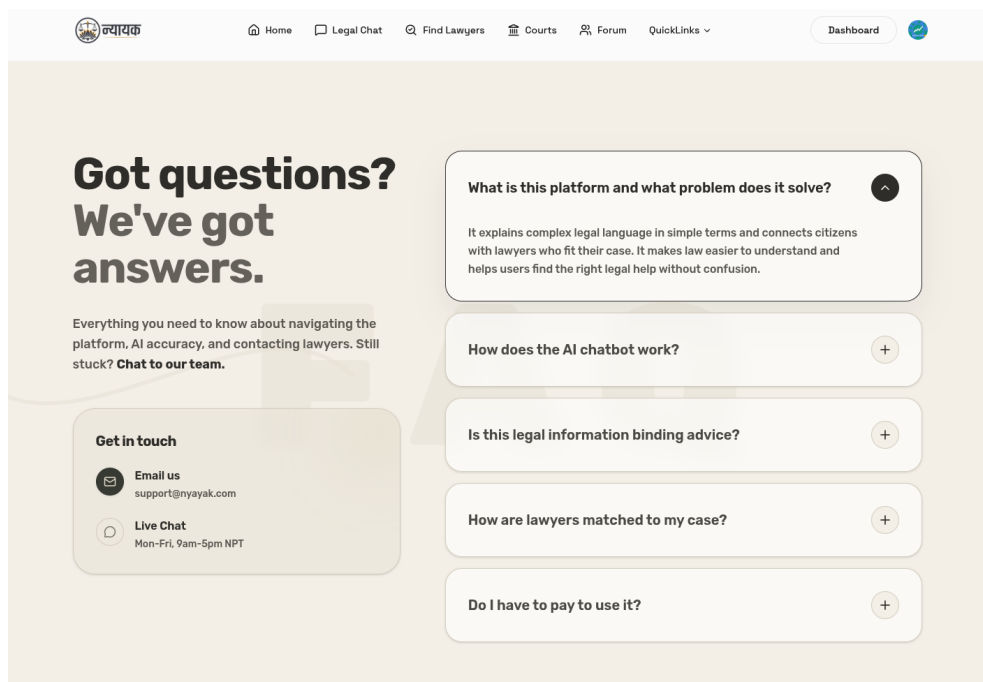
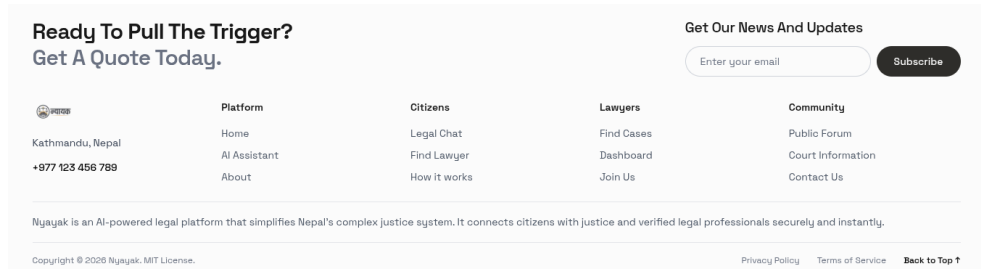


Figure A.12: FAQ Section



**Figure A.13: Footer Section**

## B. Major Source Code Components

### B.1 FastAPI Backend Entry Point (backend/main.py)

```

1  import os
2  import logging
3  from fastapi import FastAPI
4  from fastapi.middleware.cors import CORSMiddleware
5  from dotenv import load_dotenv
6  from groq import Groq
7  from models import LegalQuery, LegalResponse
8  from rag_pipeline import RAGPipeline, load_constitution_pdf
9
10 # Initialize FastAPI app
11 app = FastAPI(
12     title="Synergy Legal Assistant API",
13     description="Python backend for RAG-powered legal consultation",
14     version="1.0.0"
15 )
16
17 # Enable CORS
18 app.add_middleware(
19     CORSMiddleware,
20     allow_origins=["http://localhost:3000", "http://localhost:3001"],
21     allow_credentials=True,
22     allow_methods=["*"],
23     allow_headers=["*"],
24 )

```

```

25
26 # Initialize RAG pipeline
27 constitution_docs = load_constitution_pdf()
28 rag_pipeline = RAGPipeline(constitution_docs)
29
30 @app.post("/api/ask-legal", response_model=LegalResponse)
31 async def ask_legal(request: LegalQuery):
32     """Legal consultation endpoint with RAG"""
33     query = request.query
34     relevant_docs = rag_pipeline.query(query)
35     # Process and return response
36     return {"answer": answer, "sources": sources}

```

## B.2 RAG Pipeline (backend/rag\\_pipeline.py)

```

1 class HybridVectorStore:
2     """Fast hybrid search: BM25 keyword + semantic embeddings"""
3
4     def hybrid_search(self, query: str, k: int = 3) -> List[
5         Dict]:
6         """Hybrid search combining BM25 and semantic similarity"""
7         results = {}
8
9         # BM25 keyword search (70% weight)
10        if self.bm25:
11            query_tokens = query.lower().split()
12            bm25_scores = self.bm25.get_scores(query_tokens)
13            for idx, score in enumerate(bm25_scores):
14                if score > 0.01:
15                    results[idx] = score * 0.7
16
17        # Semantic search with embeddings (30% weight)
18        if self.embeddings and self.doc_embeddings is not
19        None:
20            query_embedding = self.embeddings.encode(query)
21            similarities = cosine_similarity(
22                [query_embedding], self.doc_embeddings)[0]
23            for idx, score in enumerate(similarities):
24                if score > 0.1:

```

```

23         results[idx] = results.get(idx, 0) + (
24             score * 0.3)
25
26     sorted_results = sorted(results.items(),
27                             key=lambda x: x[1], reverse=
28                             True)[:k]
29
30     return [self.documents[idx] for idx, _ in
31             sorted_results]

```

### B.3 Convex Database Schema (convex/schema.ts)

```

1  export default defineSchema({
2      users: defineTable({
3          name: v.optional(v.string()),
4          email: v.string(),
5          clerkId: v.string(),
6          role: v.union(v.literal("user"),
7                      v.literal("lawyer"),
8                      v.literal("admin"))
9      }).index("by_clerkId", ["clerkId"]),
10
11     lawyers: defineTable({
12         userId: v.id("users"),
13         specializations: v.array(v.string()),
14         yearsOfExperience: v.number(),
15         rating: v.number(),
16         phone: v.string(),
17         bio: v.string(),
18         availableNow: v.boolean()
19     }).index("by_userId", ["userId"]),
20
21     forum_posts: defineTable({
22         title: v.string(),
23         content: v.string(),
24         authorId: v.id("users"),
25         category: v.optional(v.string()),
26         upvotes: v.number()
27     }).index("by_authorId", ["authorId"]),
28
29     forum_comments: defineTable({
30         postId: v.id("forum_posts"),

```

```

31     authorId: v.id("users"),
32     content: v.string(),
33     parentId: v.optional(v.id("forum_comments")),
34     upvotes: v.number()
35   }).index("by_postId", ["postId"])
36 });

```

#### B.4 Lawyer Recommendation Logic (convex/lawyers.ts)

```

1  export const recommend = query({
2    args: { query: v.string() },
3    handler: async (ctx, args) => {
4      const lawyerProfiles = await ctx.db.query("lawyers").
5        collect();
6      const keywords = extractKeywords(args.query);
7
8      const lawyers = await Promise.all(
9        lawyerProfiles.map(async (profile) => {
10
11          // 1. Specialization Match (20 points)
12          const exactMatch = profile.specializations.some(s =>
13            args.query.toLowerCase().includes(s.toLowerCase())
14          );
15          if (exactMatch) score += 20;
16
17          // 2. Keyword Match in Bio (2 points per hit)
18          const bioLower = profile.bio.toLowerCase();
19          keywords.forEach(k => {
20            if (bioLower.includes(k)) score += 2;
21          });
22
23          // 3. Rating Bonus (1-5 points)
24          score += (profile.rating || 0);
25
26          // 4. Experience Bonus (up to 5 points)
27          score += Math.min(profile.yearsOfExperience * 0.2, 5)
28            ;
29
30          // 5. Availability Bonus (3 points)
31          if (profile.availableNow) score += 3;

```

```

31
32     return { ...profile, score, user };
33   })
34 );
35 return lawyers.sort((a, b) => b.score - a.score);
36 }
37 });

```

## B.5 Forum PageRank and DFS (convex/forum.ts)

```

1  export const getCommentsAndTraverse = query({
2    args: { postId: v.id("forum_posts") },
3    handler: async (ctx, args) => {
4      const comments = await ctx.db.query("forum_comments")
5        .withIndex("by_postId", q => q.eq("postId", args.postId
6          ))
7        .collect();
8
9      // Build adjacency list for threading
10     const graph: Record<string, any[]> = {};
11     const roots: any[] = [];
12
13     comments.forEach((comment) => { graph[comment._id] = [];
14       });
15
16     comments.forEach((comment) => {
17       if (comment.parentId && graph[comment.parentId]) {
18         graph[comment.parentId].push(comment);
19       } else {
20         roots.push(comment);
21       }
22     });
23
24     // DFS Traversal for chronological thread reconstruction
25     const flattenTreeDFS = (nodeList: any[], depth = 0): any
26       [] => {
27       let result: any[] = [];
28       for (const node of nodeList) {
29         result.push({ ...node, depth });
30         const children = graph[node._id] || [];
31         if (children.length > 0) {

```

```

29         result = result.concat(flattenTreeDFS(children,
30             depth + 1));
31     }
32     }
33     return result;
34 };
35
36 return flattenTreeDFS(roots);
37 });

```

## B.6 Chat Interface Component (src/components/chat/ChatInterface.tsx)

```

1  const sendMessage = async (text: string) => {
2      if (!text.trim()) return;
3
4      let currentSessionId = activeSessionId;
5
6      // Create session if needed
7      if (!currentSessionId) {
8          const title = text.length > 30 ? text.substring(0, 30) :
9              text;
10         currentSessionId = await createSession({ title });
11         onSessionCreated(currentSessionId);
12     }
13
14     // Save user message
15     await addMessage({
16         sessionId: currentSessionId,
17         text,
18         sender: "user",
19     });
20
21     setLoading(true);
22     try {
23         // Call backend RAG API
24         const response = await fetch("/api/ask-legal", {
25             method: "POST",
26             headers: { "Content-Type": "application/json" },
27             body: JSON.stringify({ query: text }),
28         });

```

```
28
29     const data = await response.json();
30
31     // Save assistant response
32     await addMessage({
33         sessionId: currentSessionId,
34         text: data.answer,
35         sender: "assistant",
36     });
37 } catch (error) {
38     console.error(error);
39 } finally {
40     setLoading(false);
41 }
42 };
```

## C. Project Progress Log Report